

HansRobot Library

Python

(Common to Elfin series robots.)



Revision history

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V1.0.0.6	2023.02.23	HR5.50.4	M Revise the interpretation of HRIF_WayPointRel interface's parameter nIsUseJoint
V1.0.1.0	2023.02.23	HR5.50.3	A Added HRIF_GetErrorCodeStr interface
V1.0.2.0	2023.02.23	HR5.50.3	A Added location tracking interface HRIF_SetPoseTrackingMaxMotionLimit HRIF_SetPoseTrackingPIDParams HRIF_SetPoseTrackingTargetPos HRIF_SetPoseTrackingState HRIF_SetUpdateTrackingPose
V1.0.2.1	2023.03.21	HR5.51.0	M Revise interface HRIF_IsBlendingDone interface

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Chapter I Overview

1.1.Purpose of writing

- 1.1. Provide interface usage instructions for people using the Hans Python interface.
- 1.2. Provide reference documents for interface writing and maintenance.

1.2.Terminology Explanation

Special name	Explain
ACS	Joint coordinates, unit (°)
PCS	Spatial coordinates, unit(mm), (°)
TCP	The default tool coordinate system of the system, with initial values of 0,0,0,0,0,0
UCS	User coordinate system
Base	The system has the default user coordinate system, with the initial value of 0,0,0,0,0,0
Tool	System default tool coordinate system

1.3. Data type

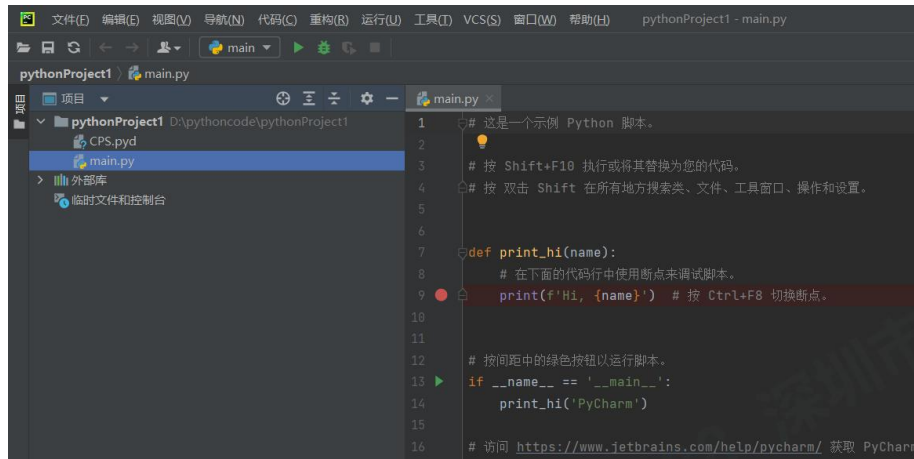
Data type	Length	Explain
bool	-	Boolean type
int	-	Integer
float	-	Floating point type
string	-	String
list		Array

Chapter II Environment build

2.1.Windows Environment build

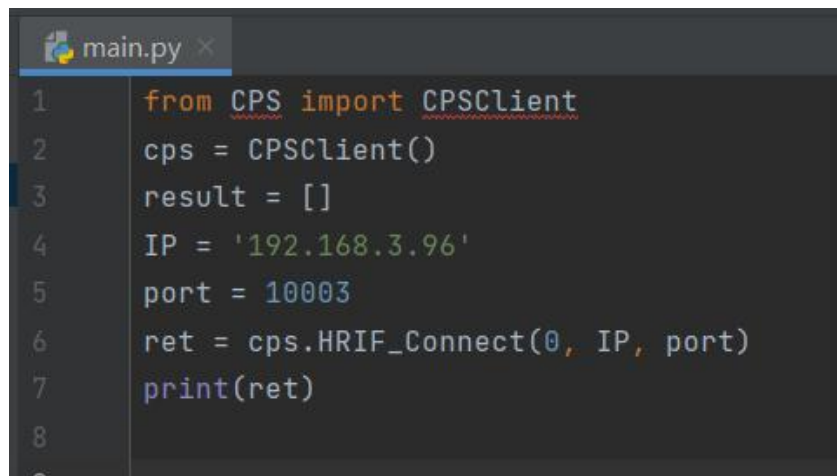
2.1.1.Create a project

Rename the package to CPS.pyd, copy it to an empty directory, and create a Python file, main.py.(Take PyCharm as an example)



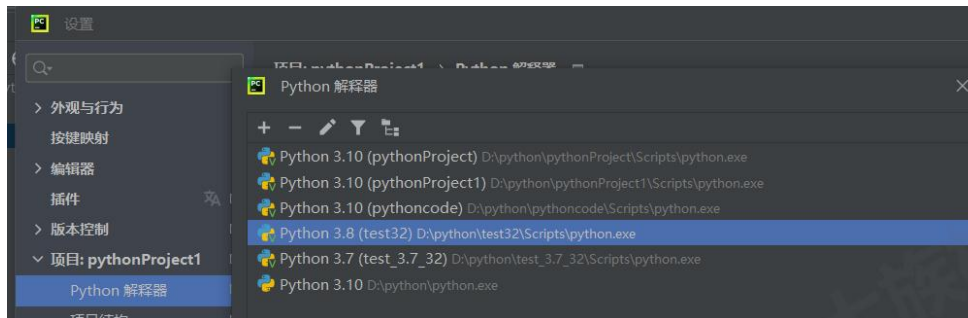
2.1.2. Quote SDK

Introduce the CPSCClient object within the main.py and try to connect.



2.1.3. Version selection

According to the package body used, select the compiler of the corresponding version. If the original package body is called CPS_3.8.9_win_32.pyd, the 32-bit Python 3.8 version should be selected.



2.1.4. Action routine

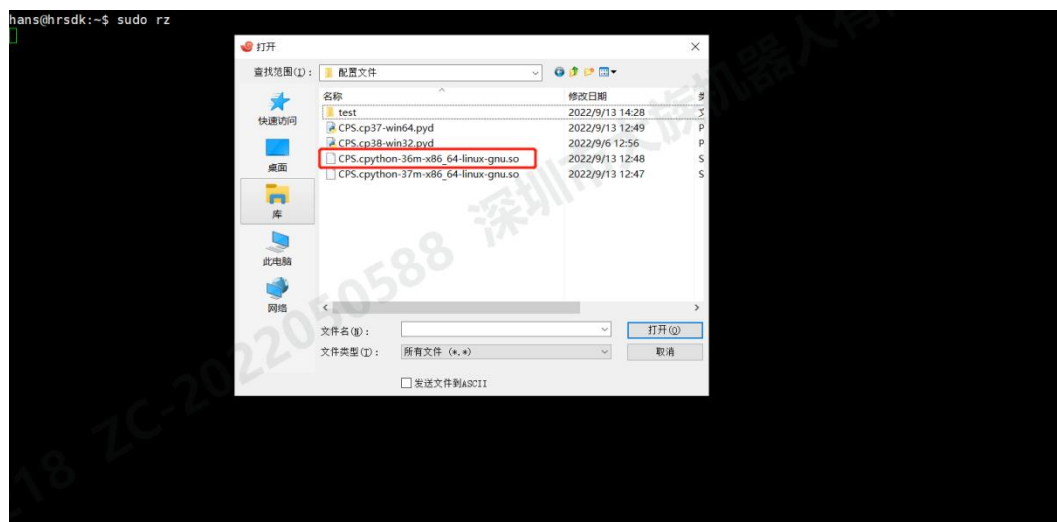
Right-click the workspace and select run code to view the terminal print below



2.2. Linux Environment build

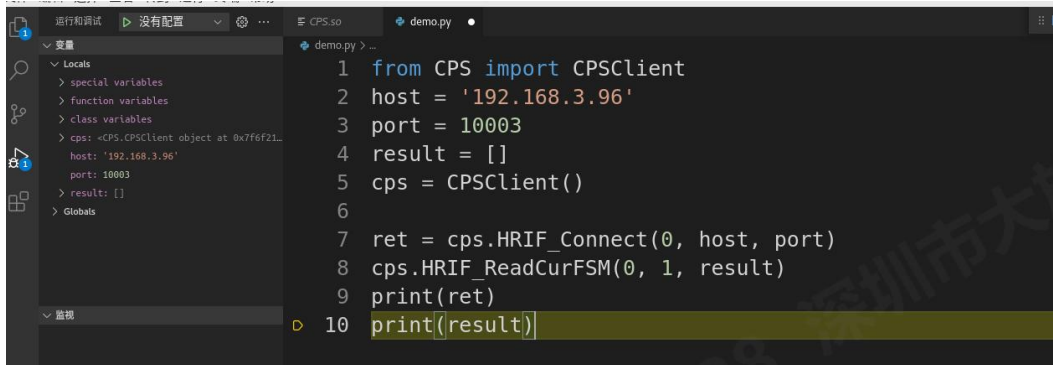
2.2.1. Upload SDK

Upload the SDK file to the system (take vscode, for example)



2.2.2. Quote SDK

Create a new folder, place the so file in it, and open it via vscode. Create a new demo.py file, introduce the CPSClnt object, and try to connect.

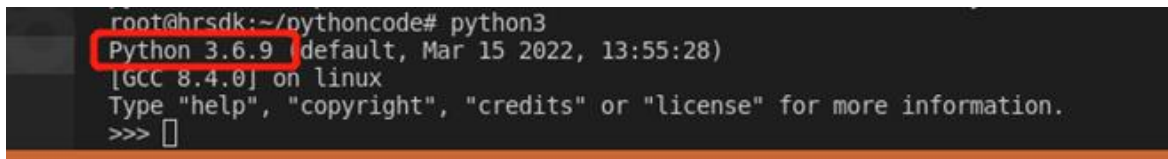


```

1 from CPS import CPSClnt
2 host = '192.168.3.96'
3 port = 10003
4 result = []
5 cps = CPSClnt()
6
7 ret = cps.HRIF_Connect(0, host, port)
8 cps.HRIF_ReadCurFSM(0, 1, result)
9 print(ret)
10 print(result)
  
```

2.2.3.Version selection

According to the SDK used, select the compiler if the SDK named CPS_python3_Linux.so means that any version of Python 3 should be selected.

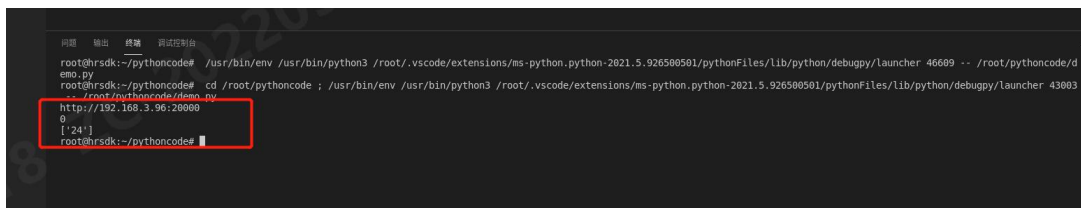


```

root@hrsdk:~/pythoncode# python3
Python 3.6.9 [default, Mar 15 2022, 13:55:28]
[GCC 8.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
  
```

2.2.4. Action routine

Run the program to view the terminal print.



```

root@hrsdk:~/pythoncode# /usr/bin/env /usr/bin/python3 /root/.vscode/extensions/ms-python.python-2021.5.926500501/pythonFiles/lib/python/debugpy/launcher 46609 -- /root/pythoncode/d
emo.py
root@hrsdk:~/pythoncode# cd /root/pythoncode ; /usr/bin/env /usr/bin/python3 /root/.vscode/extensions/ms-python.python-2021.5.926500501/pythonFiles/lib/python/debugpy/launcher 43003
http://192.168.3.96:20000
0
[ 24 ]
root@hrsdk:~/pythoncode#
  
```

Chapter III Interface Description

3.1. Initialise

3.1.1.HRIF_Connect

3.1.1.1.Description: Connect to the robot server.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
IP	Controller IP address	string	-	Controller IP address, IP address definitions according to the actual settings
nPort	Controller port	int		Controller port, generally 10003

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : The function call was successful nRet>0: Returns the error code of the call failure

3.1.1.2.Example

```
# Import into the Python SDK
from CPS import CPSCient
# Define the IP, and the port
IP = '192.168.0.10'
nPort = 10003
# Create objects
cps = CPSCient()
# Connection controller
nRet = cps.HRIF_Connect(0,IP,nPort)
```

3.1.2.HRIF_DisConnect

3.1.2.1.Description: Disconnect from the robot server.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.2.2.Example

Disconnect from the robot server

nRet = cps.HRIF_DisConnect(0)

3.1.3.HRIF_IsConnected

3.1.3.1.Description: Determine whether the controller is connected.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
bRet	Return value	bool	False,True	False : The controller is not connected True : The controller is connected

3.1.3.2.Example

Whether the controller is connected or not

nRet = cps.HRIF_IsConnected(0)

3.1.4.HRIF_ShutdownRobot

3.1.4.1.Description: Controller power off (off robot power supply, system shutdown)

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.4.2.Example

The controller is off power

nRet = cps.HRIF_ShutdownRobot(0)

3.1.5.HRIF_Connect2Box

3.1.5.1.Description: Connect the controller power box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.5.2.Example

Connect the controller power box

nRet = cps.HRIF_Connect2Box(0)

3.1.6.HRIF_Electrify

3.1.6.1.Description: Power up the robot.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.6.2.Example

Robot on electricity

nRet = cps.HRIF_Electrify(0)

3.1.7.HRIF_Blackout

3.1.7.1.Description: The robot is powered off.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.7.2.Example

The robot power

nRet = cps.HRIF_Blackout(0)

3.1.8.HRIF_Connect2Controller

3.1.8.1.Description: Connect the controller, start the main station, initialize the slave station, configure the configuration parameters, check the configuration, and jump to the disenabled state after completion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.8.2.Example

Connected to the robot

nRet = cps.HRIF_Connect2Controller(0)

3.1.9.HRIF_IsSimulateRobot

3.1.9.1.Description: Is it a simulated robot not.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Output variable

Output variable	Name	Data type	Valid range	Content
nSimulateRobot	Whether to simulate the robot	int	0/1	0 : Real robot 1 : Dummy robot

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.9.2.Example

Defines whether it is a simulated robot

nSimulateRobot = 0

Determine whether it is a simulated robot

nRet = cps.HRIF_IsSimulateRobot(0, nSimulateRobot)

3.1.10.HRIF_IsControllerStarted

3.1.10.1.Description: Whether the controller is started is complete.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Enter the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Whether to start completion	string	0/1	0: Unstarted 1: Started

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.10.2.Example

Define the return value-empty list

result = []

Determine whether the robot startup is complete

nRet = cps.HRIF_IsControllerStarted(0, result)

3.1.11.HRIF_GetErrorCodeStr

3.1.11.1.Description: Get the error code interpretation.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
nErrorCode	Error code	int	-	Error code

✓ Output variable

Output variable	Name	Data type	Valid range	Content
strErrorMsg	Error code description	StringData	-	A specific description of the error code

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0: Returns that the function call was successful nRet>0: Returns the error code of the call failure

3.1.11.2.Example

```
// Error code
nErrorCode = 20018
//Define the Return value list
result = []
// Call the interface
nRet = cps.HRIF_GetErrorCodeStr(0,nErrorCode,result);
// Get the error code description
```

3.1.12.HRIF_ReadVersion

3.1.12.1.Description: Read the controller version number.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	CPS edition	string	-	CPS edition number
result[1]	Controller version	string	-	Controller version number
result[2]	Electric box version	string	0~4	Box version number: 0: Simulated electric box 1~4: Version number of the electric box
result[3]	Control board firmware version	string	-	Control board firmware version
result[4]	Control board firmware version	string	-	Control board firmware version
result[5]	Algorithm version	string	-	Algorithm version
result[6]	Firmware version	string	-	Firmware version
result[7]	Software release	string	-	Software release

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.12.2.Example

Define the Return value-empty list

result = []

Get version information

nRet = cps.HRIF_ReadVersion(0,0, result)

3.1.13.HRIF_ReadRobotModel

3.1.13.1.Description: Read the robot type.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Robot type	string	-	Robot type

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.1.13.2.Example

#Define the Return value-empty list

result = []

Read the robot type

nRet = cps.HRIF_ReadRobotModel (0,0, result)

3.2.Axle group control instructions

3.2.1.HRIF_GrpEnable

3.2.1.1.Description: The Robot enables the command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.1.2.Example

Robotic enabling

nRet = cps.HRIF_GrpEnable(0,0)

3.2.2.HRIF_GrpDisable

3.2.2.1.Axle group control instructions

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.2.2.Example

Robot to enable

```
nRet = cps.HRIF_GrpDisable(0,0)
```

3.2.3.HRIF_GrpReset

3.2.3.1.Description: The robot-reset command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.3.2.Example

Robot reset

nRet = cps.HRIF_GrpReset(0,0)

3.2.4.HRIF_GrpStop

3.2.4.1.Description: The robot stops the motion command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.4.2.Example

The robot stops moving

nRet = cps.HRIF_GrpStop(0,0)

3.2.5. HRIF_Grppinterrupt

3.2.5.1.Description: The robot will pause the motion command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.5.2.Example

The robot is suspended in motion

nRet = cps.HRIF_Grppinterrupt(0,0)

3.2.6. HRIF_GrpContinue

3.2.6.1.Description: The robot continues the motion command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.6.2.Example

The robot continues to motion

nRet = cps.HRIF_GrpContinue(0,0)

3.2.7.HRIF_GrpCloseFreeDriver

3.2.7.1.Description: Close the free drive.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.7.2.Example

The robot turns off the free drive

nRet = cps.HRIF_GrpCloseFreeDriver(0,0)

3.2.8.HRIF_GrpOpenFreeDriver

3.2.8.1.Description: Open the free drive.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.2.8.2.Example

The robot turns on the free drive

nRet = cps.HRIF_GrpOpenFreeDriver(0,0)

3.3.Script Control Directive

3.3.1.HRIF_RunFunc

3.3.1.1.Description: Run the specified script function.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
strFuncName	Function name	string	-	Specify the script function name, the function of the presentation interface
param	Parameter list	string	-	List of parameters
result	Return value	list		Pass into the empty list, result = []

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.3.1.2.Example

```
# Specifies the run function Func_1
strFuncName = "Func_1"
# Define input parameters
param = [1,2,3]
# Define the Return value-empty list
result = [ ]
# Run the function Func _ 1
nRet = cps.HRIF_RunFunc(0,0, strFuncName, param,result)
```


3.3.2.HRIF_StartScript

3.3.2.1.Description: Execute the script Main function, call and execute the compiled script file Finc _ main function.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.3.2.2.Example

Run the script Main function

nRet = cps.HRIF_StartScript(0)

3.3.3.HRIF_StopScript

3.3.3.1.Description: Stop the script and stop the user page after calling is executing the script file.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.3.3.2.Example

Stop the currently running script

nRet = cps.HRIF_StopScript(0,0)

3.3.4.HRIF_PauseScript

3.3.4.1.Description: Pause the script and then the user page is executing the script file.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.3.4.2.Example

Pause the currently running script

nRet = cps.HRIF_PauseScript(0)

3.3.5.HRIF_ContinueScript

3.3.5.1.Description: Continue the script, call after the user page paused script file, not pause state return 20018 error.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.3.5.2.Example

Continue running the currently suspended script

nRet = cps.HRIF_ContinueScript(0)

3.4.Electric box control instructions

3.4.1.HRIF_ReadBoxInfo

3.4.1.1.Description: Read the box information.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Electric box connection status	string	0/1	0: Not connected 1: Connected
result[1]	The 48V voltage state	string	0/1	0: Voltage is <36V 1: Voltage> 36V
result[2]	48V output voltage value	string	0~53	48V output voltage value
result[3]	48V output current value	string	-	48V output current value
result[4]	Remote emergency stop state	string	0/1	0: No signal 1: Have a signal
result[5]	Three-stage button status	string	0/1	0: No signal 1: Have a signal

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.1.2.Example

Define the return value-empty list

result = []

Read the box information

nRet = cps.HRIF_ReadBoxInfo(0,0, result)

3.4.2.HRIF_ReadBoxCI

3.4.2.1.Description: Read the box control digital input status.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	Control the digital input bits	int	0~7	Control the digital input bit index

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Control the number input corresponding bit state	string	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.2.2.Example

Defines the bits that you need to read

nBit = 0

Define the return value-empty list

result = []

Read the power box nBit bit CI status

nRet = cps.HRIF_ReadBoxCI (0,nBit,result)

3.4.3.HRIF_ReadBoxDI

3.4.3.1.Description: Read the universal digital input status of the power box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	Universal digital input bit	int	0~7	Universal digital input bit index

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Common digit input corresponds to the bit states	string	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.3.2.Example

Defines the bits that you need to read

nBit = 0

Define the return value-empty list

result = []

Read the box nBit bit DI status

nRet = cps.HRIF_ReadBoxDI (0, nBit,result)

3.4.4.HRIF_ReadBoxCO

3.4.4.1.Read the box nBit bit DI status

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	Control the digital output bits	int	0~7	Control the digital output bits

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Control of the digital output corresponds to the bit status	int	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.4.2.Example

Defines the bits that you need to read

nBit = 0

Define the return value-empty list

result = []

Read the power box nBit bit CO status

nRet = cps.HRIF_ReadBoxCO (0, nBit, result)

3.4.5.HRIF_ReadBoxDO

3.4.5.1.Description: Read the universal digital output status of the power box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	Universal digital output bits	int	0~7	Universal digital output bits

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Universal digital output corresponds to the bit states	int	0/1	Universal digital output bit index

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.5.2.Example

Defines the bits that you need to read

nBit = 0

Define the return value-empty list

result = []

Read the nBit bit DO state of the power box

nRet = cps.HRIF_ReadBoxDO (0, nBit, result)

3.4.6.HRIF_ReadBoxAI

3.4.6.1.Description: Read the analog input value of the power box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	Analog input channel	int	0/1	0: analog quantity channel 0 1: analog quantity channel 1

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Corresponding analog quantity channel values	float	0~20	Current mode: 4-20 mA Voltage mode: 0-10V

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.6.2.Example

Defines the bits that you need to read

nBit = 0

Define the return value-empty list

result = []

Read the power box nBit channel AI value

nRet = cps.HRIF_ReadBoxAI (0, nBit, result)

3.4.7.HRIF_ReadBoxAO

3.4.7.1.Description: Read the analog output value of the power box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	analog output channel	int	0/1	0: analog quantity channel 0 1: analog quantity channel 1

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Corresponding analog quantity channel patterns	string	0/1	0: Voltage mode 1: Current mode:
result[1]	Corresponding analog quantity channel values	string	0~20	Current mode: 4-20 mA Voltage mode: 0-10V

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.7.2.Example

Defines the bits that you need to read

nBit = 0

Define the return value-empty list

result = []

Read the power box nBit channel AO value

nRet = cps.HRIF_ReadBoxAO(0, nBit, result)

3.4.8.HRIF_SetBoxCO

3.4.8.1.Description: Set the digital output status of the control.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
nBit	Control the digital output bits	int	0~7	Control the digital output bits index
nVal	Control of the digital output status	int	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.8.2.Example

Define the setting state

nVal = 0

Defines the bits that need to be set

Bit = 0

Set the nBit bit CO state of the power box to nVal

nRet = cps.HRIF_SetBoxCO (0, nBit, nVal)

3.4.9.HRIF_SetBoxDO

3.4.9.1.Description: Set the universal digital output status of the power box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
nBit	Universal digital output bits	int	0~7	Universal digital output bit index
nVal	Universal digital output status	int	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.9.2.Example

Define the setting state

nVal = 0

Defines the bits that need to be set

nBit = 0

Set the nBit bit DO state of the power box to nVal

nRet = cps.HRIF_SetBoxDO(0, nBit, nVal)

3.4.10.HRIF_SetBoxAOMode

3.4.10.1.Description: Set the box analog output mode.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
nBit	Simulation channel	int	0/1	Simulation channel
nVal	model	int	1/2	1: Voltage mode 2: Current mode

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.10.2.Example

Define the setting value

nVal = 0

Define the channel that you need to set up

nBit = 0

Set the box nBit channel AO mode to nVal

nRet = cps.HRIF_SetBoxAOMode (0,nBit , nVal)

3.4.11.HRIF_SetBoxAOVal

3.4.11.1.Description: Set the output value of the box.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
nBit	Simulation channel	int	0/1	Simulation channel
dVal	The corresponding simulated quantity value	float	0~20	Current mode: 4-20 mA Voltage mode: 0-10V
nMode	Model	int	1/2	1: Voltage mode 2: Current mode

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.11.2.Example

Define the setting value

dVal = 0

Define the channel that you need to set up

nBit = 0

The current mode of the read

nMode = 0

Set the box nBit channel AO value to dVal

nRet = cps.HRIF_SetBoxAOVal(0,nBit, dVal, nMode)

3.4.12.HRIF_SetEndDO

3.4.12.1.Description: Set the terminal digital output status.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nBit	End-point digital output bits	int	0~3	Universal digital output bit index with only 4 DO
nVal	End-end digital output status	int	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.12.2.Example

Define the setting value

nVal = 0

Defines the bits that need to be set

nBit = 0

Sets the terminal nBit bit DO status

nRet = cps.HRIF_SetEndDO(0,0,nBit, nVal)

3.4.13.HRIF_ReadEndDI

3.4.13.1.Description: Read the terminal number input status.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nBit	End-digital input bit	int	0~2	End numbers are input into the corresponding bit states
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
reuslt[0]	End numbers are input into the corresponding bit states	string	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.13.2.Example

Define the return value-empty list

result = []

Defines the bits that you need to read

nBit = 0

Read the terminal nBit bit DI state

nRet = cps.HRIF_ReadEndDI(0,0, nBit, result)

3.4.14.HRIF_ReadEndDO

3.4.14.1.Description: Read the end digital output status.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	End-point digital output bits	int	0~2	End digital output corresponds to the bit index

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	End digital output corresponds to the bit state	string	0/1	0: Low level 1: High level

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.14.2.Example

Define the return value-empty list

result = []

Defines the bits that you need to read

nBit = 0

Read the terminal nBit bit DO state

nRet = cps.HRIF_ReadEndDO(0,0, nBit, result)

3.4.15.HRIF_ReadEndAI

3.4.15.1.Description: Read the end analog input value.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nBit	analog input channel	int	0/1	0: analog quantity channel 0 1: analog quantity channel 1

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Corresponding analog quantity channel values	string	0~20	Current mode: 4 ~ 20 mA Voltage mode: 0-10V

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.15.2.Example

Define the return value-empty list

result = []

Defines the bits that you need to read

nBit = 0

Read the terminal nBit channel AI values

nRet = cps.HRIF_ReadEndAI(0,0, nBit, result)

3.4.16.HRIF_ReadEndBTN

3.4.16.1.Description: Read the status of the end button. According to the type of end, the status will have different meanings.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Key 1 state	string	0/1	0: Loosen 1: Press
result[1]	Key 2 state	string	0/1	0: Loosen 1: Press
result[2]	Key 3 state	string	0/1	0: Loosen 1: Press
result[3]	Key 4 state	string	0/1	0: Loosen 1: Press

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.4.16.2.Example

Define the return value-empty list

result = []

Read the terminal nBit channel AI values

nRet = cps.HRIF_ReadEndBTN(0,0, result)

3.5.State reading and setting instructions

3.5.1.HRIF_SetOverride

3.5.1.1.Description: Set the speed ratio.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dOverride	Velocity ratir	float	0.01~1	Velocity ratir that needs to be set(0.01~1)

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.1.2.Example

Velocity ratir that needs to be set

dOverride = 0.5

Set up current Velocity ratir

nRet = cps.HRIF_SetOverride(0,0, dOverride)

3.5.2.HRIF_SetToolMotion

3.5.2.1.Description: Turn the Tool coordinate system motion mode on or off.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nState	status	int	0/1	0: Open 1: Close

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.2.2.Example

Tool to be set, motion state

nState = 1

Set the Tool, the motion state

nRet = cps.HRIF_SetToolMotion(0,0,nState)

3.5.3.HRIF_SetPayload

3.5.3.1.Description: Set the current load parameters.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMass	质量	float	0~允许最大负载	质量, 范围为 0 到当前允许最大负载, unit[kg]
dX	Centroid X direction offset	float	>0	Centroid X direction offset,unit [mm]
dY	Centroid Y direction offset	float	>0	Centroid Y direction offset,unit [mm]
dZ	Centroid Z direction offset	float	>0	Centroid Z direction offset,unit [mm]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.3.2.Example

Load mass needs to be set

dMass = 0.5

The offset in the X direction needs to be set

dX= 0.5

The offset in the Z direction needs to be set

dY= 0.5

The offset in the Z direction needs to be set

dZ= 0.5

Set the current load

nRet = cps.HRIF_SetPayload(0,0,dMass, dX, dY, dZ)

3.5.4.HRIF_SetJointMaxVel

3.5.4.1.Description: Set the maximum movement speed of the joint.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dJ1	J1 axis maximum speed	float	0~200	J1 axis maximum speed, unit[°/s]
dJ2	J2 axis maximum speed	float	0~200	J2 axis maximum speed, unit[°/s]
dJ3	J3 axis maximum speed	float	0~200	J3 axis maximum speed, unit[°/s]
dJ4	J4 axis maximum speed	float	0~200	J4 axis maximum speed, unit[°/s]
dJ5	J5 axis maximum speed	float	0~200	J5 axis maximum speed, unit[°/s]
dJ6	J6 axis maximum speed	float	0~200	J6 axis maximum speed, unit[°/s]

* Note: Joint speed Valid range needs to refer to the specific model.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.4.2.Example

Definition requires a maximum joint speed

Joint = [180, 180, 180, 180, 180, 180]

Set the maximum joint speed

nRet = cps.HRIF_SetJointMaxVel(0,0,Joint)

3.5.5.HRIF_SetJointMaxAcc

3.5.5.1.Description: Set the maximum movement acceleration of the joints.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dJ1	J1 axis maximum acceleration	float	0~720	J1 axis maximum acceleration, unit[°/s ²]
dJ2	J2 axis maximum acceleration	float	0~720	J2 axis maximum acceleration, unit[°/s ²]
dJ3	J3 axis maximum acceleration	float	0~720	J3 axis maximum acceleration, unit[°/s ²]
dJ4	J4 axis maximum acceleration	float	0~720	J4 axis maximum acceleration, unit[°/s ²]
dJ5	J5 axis maximum acceleration	float	0~720	J5 axis maximum acceleration, unit[°/s ²]
dJ6	J6 axis maximum acceleration	float	0~720	J6 axis maximum acceleration, unit[°/s ²]

* Note: Joint acceleration of Valid range needs to refer to the specific model.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.5.2.Example

Define the maximum joint acceleration that needs to be set

JointAcc = [20, 20, 20, 20, 20, 20,]

Set the maximum joint acceleration

nRet = cps.HRIF_SetJointMaxAcc(0,0,JointAcc)

3.5.6.HRIF_SetLinearMaxVel

3.5.6.1.Description: Set the maximum speed of linear motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMaxVel	Maximum line speed	float	0~3000	Maximum linear speed, default [500], unit [mm/s]

* Note: The linear speed Valid range needs refers to the specific model.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.6.2.Example

Definition requires setting a straight line maximum speed

dMaxVel = 100

Set the maximum rectilinear speed

nRet = cps.HRIF_SetLinearMaxVel(0,0,dMaxVel)

3.5.7.HRIF_SetLinearMaxAcc

3.5.7.1.Description: Set the maximum acceleration of linear movement.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMaxAcc	Maximum rectilinear acceleration	float	0~2500	Maximum linear speed, default [2500], unit [mm/s ²]

* Note: Linear acceleration Valid range needs to refer to the specific model.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.7.2.Example

Definition requires setting a straight line maximum speed

dMaxAcc = 2500

Set the maximum rectilinear speed

nRet = cps.HRIF_SetLinearMaxAcc(0,0,dMaxAcc)

3.5.8.HRIF_SetMaxAcsRange

3.5.8.1.Description: Set the maximum range of motion of the joint.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMaxJ1- dMaxJ6	Maximum joint range	float	-	dMaxJ1:Maximum range of motion for Joint 1, unit[°] dMaxJ2:Maximum range of motion for Joint 2, unit[°] dMaxJ3:Maximum range of motion for Joint 3, unit[°] dMaxJ4:Maximum range of motion for Joint 4, unit[°] dMaxJ5:Maximum range of motion for Joint 5, unit[°] dMaxJ6:Maximum range of motion for Joint 6, unit[°]
dMinJ1- dMinJ6	Joint minimum range	float	-	dMinJ1:Minimum range of motion for Joint 1, unit[°] dMinJ2:Minimum range of motion for Joint 2,unit[°] dMinJ3:Minimum range of motion for Joint 3,unit[°] dMinJ4:Minimum range of motion for Joint 4, unit[°] dMinJ5:Minimum range of motion for Joint 5,unit[°] dMinJ6:Minimum range of motion for Joint 6,unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful

				nRet>0 : Returns the error code of the call failure
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3.5.8.2.Example

Define the maximum range of motion of the joint to be set

dMax = [360, 360, 360, 360, 360, 360]

Define the minimum range of joint motion to be set

dMin = [-360, -360, -360, -360, -360, -360]

Set the range of joint motion

nRet = cps.HRIF_SetMaxAcsRange (0,0,dMax, dMin)

3.5.9.HRIF_SetMaxPcsRange

3.5.9.1.Description: Set the maximum motion range of the space.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMax	XYZ maximum magnitude	float	-	XYZ maximum magnitude, unit[mm]
dMin	XYZ Minimum magnitude	float	-	XYZ Minimum Magnitude, unit[mm]
dUcs	Based on user coordinates	float	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.9.2.Example

Defines the maximum spatial range of motion that needs to be set

dMax = [360, 360, 360]

Deffines the spatial minimum motion range to be set

dMin = [-360, -360, -360]

Define the user coordinate variable

dUcs = [0, 0, 0, 0, 0, 0]

Set the range of joint motion

nRet = cps.HRIF_SetMaxPcsRange (0,0,dMax, dMin, dUcs)

3.5.10.HRIF_ReadOverride

3.5.10.1.Description: Read Velocity ratir.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable :

Output variable	Name	Data type	Valid range	Content
result[0]	Velocity ratir	string	0.01~1	Velocity ratir of current system of: (0.01~1)

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.10.2.Example

Define the return value-empty list

result = []

Read the maximum joint speed

nRet = cps.HRIF_ReadOverride(0,0, result)

3.5.11.HRIF_ReadJointMaxVel

3.5.11.1.Description: Read the maximum movement speed of the joint.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable :

Output variable	Name	Data type	Valid range	Content
result[0]	J1 axis maximum speed	string	0~200	J1 axis maximum speed, unit[°/s]
result[1]	J2 axis maximum speed	string	0~200	J2 axis maximum speed, unit[°/s]
result[2]	J3 axis maximum speed	string	0~200	J3 axis maximum speed, unit[°/s]
result[3]	J4 axis maximum speed	string	0~200	J4 axis maximum speed, unit[°/s]
result[4]	J5 axis maximum speed	string	0~200	J5 axis maximum speed, unit[°/s]
result[5]	J6 axis maximum speed	string	0~200	J6 axis maximum speed, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.11.2.Example

Define the return value-empty list

result = []

Read the maximum joint speed

nRet = cps.HRIF_ReadJointMaxVel(0,0, result)

3.5.12.HRIF_ReadJointMaxAcc

3.5.12.1.Description: Read the maximum movement acceleration of the joint.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable :

Output variable	Name	Data type	Valid range	Content
result[0]	J1 axis maximum acceleration	string	0~720	J1axis maximum acceleration, unit[°/s ²]
result[1]	J2 axis maximum acceleration	string	0~720	J2axis maximum acceleration, unit[°/s ²]
result[2]	J3 axis maximum acceleration	string	0~720	J3axis maximum acceleration, unit[°/s ²]
result[3]	J4 axis maximum acceleration	string	0~720	J4axis maximum acceleration, unit[°/s ²]
result[4]	J5 axis maximum acceleration	string	0~720	J5axis maximum acceleration, unit[°/s ²]
result[5]	J6 axis maximum acceleration	string	0~720	J6axis maximum acceleration, unit[°/s ²]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.12.2.Example

Define the return value-empty list

result = []

#Read the maximum joint acceleration

nRet = cps.HRIF_ReadJointMaxAcc(0,0, result)

3.5.13.HRIF_ReadJointMaxJerk

3.5.13.1.Description: Read the maximum movement acceleration of the joint.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable :

Output variable	Name	Data type	Valid range	Content
result[0]	J1 axis maximum jerk	string	0~720	J1axis maximum jerk , unit[°/s³]
result[1]	J2 axis maximum jerk	string	0~720	J2axis maximum jerk , unit[°/s³]
result[2]	J3 axis maximum jerk	string	0~720	J3axis maximum jerk , unit[°/s³]
result[3]	J4 axis maximum jerk	string	0~720	J4axis maximum jerk , unit[°/s³]
result[4]	J5 axis maximum jerk	string	0~720	J5axis maximum jerk , unit[°/s³]
result[5]	J6 axis maximum jerk	string	0~720	J6axis maximum jerk , unit[°/s³]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.13.2.Example

Define the return value-empty list

result = []

Read the maximum joint jerk

nRet = cps.HRIF_ReadJointMaxJerk(0,0, result)

3.5.14.HRIF_ReadLinearMaxSpeed

3.5.14.1.Description: Read the maximum speed parameter of the linear motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Maximum line speed	string	0~3000	Maximum line speed, default[500], unit[mm/s]
result[1]	Maximum rectilinear acceleration	string	0~2500	Maximum line speed, default[2500], unit[mm/s ²]
result[2]	Maximum linear jerk	string	0~100000	Maximum line speed, default[100000], unit[mm/s ³]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.14.2.Example

Define the return value-empty list

result = []

Read the maximum rectilinear speed

nRet = cps.HRIF_ReadLinearMaxSpeed(0,0, result)

3.5.15.HRIF_ReadEmergencyInfo

3.5.15.1.Description: Read the emergency stop information.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Urgent stop error	string	0/1	There are two emergency stop circuits. When the two signals are different, the emergency stop circuit is considered wrong, and the value is 1.
result[1]	Urgent stop signal	string	0/1	In the emergency stop occurs, the power supply of 48V output to the body will be broken, but the power supply from 220V to 48V will not be broken.
result[2]	Safety light screen error	string	0/1	There are two security light screen circuits. When the two signals are different, the security light screen circuit is considered to be wrong, and the value is 1.
result[3]	Safe light screen signal	string	0/1	Safety light screen signal. When the safety light screen occurs, the robot will stop moving and no longer accept the movement command, and the power supply of the body will not be broken.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.15.2.Example

Define the return value-empty list

result = []

Read the maximum rectilinear speed

nRet = cps.HRIF_ReadEmergencyInfo(0, result)

3.5.16.HRIF_ReadRobotState

3.5.16.1.Description: Read the current robot status flag.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	State in motion	string	0/1	0: The robot is not in the motion state 1: the robot in motion
result[1]	Has made the energy state	string	0/1	0: The robot is not enabled 1: The robot is enabled
result[2]	Error condiction	string	0/1	0: No error has occurred 1: An error has occurred
result[3]	Error code	string	>=0	Report error code
result[4]	Error axis ID	string	0~5	Error axis ID
result[5]	Lock state	string	0/1	Lock lock state (shaft drop due by gravity after brake loosening) 0: Lock 1: Pine gate
result[6]	Suspended state	string	0/1	0: Not in the paused state 1: In used
result[7]	Urgent stop state	string	0/1	0: Not in the emergency stop state 1: It is already in an emergency stop state
result[8]	Safe light screen status	string	0/1	0: Not in the safe light screen state 1: It is already in the safe light screen state
result[9]	Already on the state	string	0/1	0: Not powered on state 1: Power-on state
result[10]	Electric box connection status	string	0/1	0: Not powered on state 1: Power-on state
result[11]	Road point motion state	string	0/1	0: In the road point motion 1: Route point movement is completed
result[12]	In place	string	0/1	0: The actual position of the robot has not yet moved to the command position 1: Movement in place (basically no error between command and actual position)

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.16.2.Example

Define the return value-empty list

result = []

Read the state

nRet = cps.HRIF_ReadRobotState(0,0, result)

3.5.17.HRIF_ReadCurWaypointID

3.5.17.1.Description: Read the WayPoint current motion ID number.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	current ID	string	-	RoadPoint Current Motion ID, with WayPoint, MoveJ, MoveL sets the same road point ID in MoveC

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.17.2.Example

Define the return value-empty list

result = []

Read the point ID

nRet = cps.HRIF_ReadCurWaypointID(0,0, result)

3.5.18.HRIF_ReadAxisErrorCode

3.5.18.1.Description: Read the error code.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	The current error code	string	>=0	Current error code, with no error code, =0
result[1]	J1 axis error code	string	>=0	J1 axis error code, with no error code =0
result[2]	J2 axis error code	string	>=0	J2 axis error code, with no error code, =0
result[3]	J3 axis error code	string	>=0	J3 axis error code, with no error code, =0
result[4]	J4 axis error code	string	>=0	J4 axis error code, with no error code, =0
result[5]	J5 axis error code	string	>=0	J5 axis error code, with no error code, =0
result[6]	J6 axis error code	string	>=0	J6 axis error code, with no error code, =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.18.2.Example

Define the return value-empty list

result = []

Read the error code

nRet = cps.HRIF_ReadAxisErrorCode(0,0, result)

3.5.19.HRIF_ReadCurFSM

3.5.19.1.Description: Read the current state machine.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Current state machine	string	>=0	Status machine, you can refer to the status machine list

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.19.2.Example

Define the return value-empty list

result = []

Read the current state machine

nRet = cps.HRIF_ReadCurFSM(0,0, result)

3.5.20.HRIF_ReadPointByName

3.5.20.1.Description: Read the point information according to the point Name.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
pointName	Point name	string	-	Point name
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint coordinates	string	-	dJ1: Joint1 coordinate, unit[°] dJ2: Joint2 coordinate, unit[°] dJ3: Joint3 coordinate, unit[°] dJ4: Joint4 coordinate, unit[°] dJ5: Joint5 coordinate, unit[°] dJ6: Joint6 coordinate, unit[°]
result[6]-result[11]	Decal coordinates	string	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
result[12]-result[17]	Current tool coordinates	string	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
result[18]-result[23]	Current user coordinates	string	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm]

				dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
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✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.20.2.Example

Define the return value-empty list

result = []

Read the current actual location information

nRet = cps.HRIF_ReadPointByName(0,0,'Point_1', result)

Read the joint position variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

Read the spatial position variable

dX = float(result[6]) dY = float(result[7]) dZ = float(result[8])

dRx = float(result[9]) dRy = float(result[10]) dRz = float(result[11])

Read the tool coordinate variable

dTcp_X = float(result[12]) dTcp_Y = float(result[13]) dTcp_Z = float(result[14])

dTcp_Rx = float(result[15]) dTcp_Ry = float(result[16]) dTcp_Rz = float(result[17])

Read the user coordinate variable

dUcs_X = float(result[18]) dUcs_Y = float(result[19]) dUcs_Z = float(result[20])

dUcs_Rx = float(result[21]) dUcs_Ry = float(result[22]) dUcs_Rz = float(result[23])

3.5.21.HRIF_ReadCurFSMFromCPS

3.5.21.1.Description: Read the current state machine.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Current state machine	int	>=0	Status machine, you can refer to the status machine list

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.21.2.Example

Define the return value-empty list

result = []

Read the current state machine

nRet = cps.HRIF_ReadCurFSMFromCPS(0,0, result)

3.5.22.HRIF_ReadRobotFlags

3.5.22.1.Description: Read the current robot status flag.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	State in motion	string	0/1	0: The robot is not in the motion state 1: the robot in motion
result[1]	Has made the energy state	string	0/1	0: The robot is not enabled 1: The robot is enabled
result[2]	Error condiction	string	0/1	0: No error has occurred 1: An error has occurred
result[3]	Error code	string	>=0	Report error code
result[4]	Error axis ID	string	0~5	Error axis ID
result[5]	Lock state	string	0/1	Lock lock state (shaft drop due by gravity after brake loosening) 0: Lock 1: Pine gate
result[6]	Suspended state	string	0/1	0: Not in the paused state 1: In used
result[7]	Road point motion state	string	0/1	0: In the road point motion 1: Route point movement is completed

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.5.22.2.Example

Define the return value-empty list

result = []

Read the state

nRet = cps.HRIF_ReadRobotFlags(0,0, result)

3.6.Position, speed, and current reading command

3.6.1.HRIF_ReadActPos

3.6.1.1.Description: Read the current actual location information.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint coordinates	string	-	dJ1: Joint1 coordinate, unit[°] dJ2: Joint2 coordinate, unit[°] dJ3: Joint3 coordinate, unit[°] dJ4: Joint4 coordinate, unit[°] dJ5: Joint5 coordinate, unit[°] dJ6: Joint6 coordinate, unit[°]
result[6]-result[11]	Decal coordinates	string	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
result[12]-result[17]	Current tool coordinates	string	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
result[18]-result[23]	Current user coordinates	string	-	dX: X coordinate, unit[mm]

				dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
--	--	--	--	---

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.1.2.Example

Define the return value-empty list

result = []

Read the current actual location information

nRet = cps.HRIF_ReadActPos(0,0, result)

Read the joint position variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

Read the spatial position variable

dX = float(result[6]) dY = float(result[7]) dZ = float(result[8])

dRx = float(result[9]) dRy = float(result[10]) dRz = float(result[11])

Read the tool coordinate variable

dTcp_X = float(result[12]) dTcp_Y = float(result[13]) dTcp_Z = float(result[14])

dTcp_Rx = float(result[15]) dTcp_Ry = float(result[16]) dTcp_Rz = float(result[17])

Read the user coordinate variable

dUcs_X = float(result[18]) dUcs_Y = float(result[19]) dUcs_Z = float(result[20])

dUcs_Rx = float(result[21]) dUcs_Ry = float(result[22]) dUcs_Rz = float(result[23])

3.6.2.HRIF_ReadCmdJointPos

3.6.2.1.Description: Read the joint command position.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint coordinates	float	-	dJ1: Joint1 Command position, unit[°] dJ2: Joint2 Command position, unit[°] dJ3: Joint3 Command position, unit[°] dJ4: Joint4 Command position, unit[°] dJ5: Joint5 Command position, unit[°] dJ6: Joint6 Command position, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.2.2.Example

Define the return value-empty list

result = []

Read the joint command position

nRet = cps.HRIF_ReadCmdJointPos(0,0, result)

Read the joint command position variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

3.6.3.HRIF_ReadActJointPos

3.6.3.1.Description: Read the actual joint position.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint coordinates	float	-	dJ1: Joint1 actual location, unit[°] dJ2: Joint2 actual location, unit[°] dJ3: Joint3 actual location, unit[°] dJ4: Joint4 actual location, unit[°] dJ5: Joint5 actual location, unit[°] dJ6: Joint6 actual location, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.3.2.Example

Define the return value-empty list

result = []

Read the actual joint position

nRet = cps.HRIF_ReadActJointPos(0,0, result)

Read out the actual joint position variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

3.6.4.HRIF_ReadCmdTcpPos

3.6.4.1.Read out the actual joint position variable

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	TCP coordinate	float	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.4.2.Example

Define the return value-empty list

result = []

Read the TCP command position

nRet = cps.HRIF_ReadCmdTcpPos(0,0,result)

Read the TCP command position

dX = float(result[0]) dY = float(result[1]) dZ = float(result[2])

dRx = float(result[3]) dRy = float(result[4]) dRz = float(result[5])

3.6.5.HRIF_ReadActTcpPos

3.6.5.1.Description: Read the actual TCP position.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	TCP coordinate	float	-	dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.5.2.Example

Define the return value-empty list

result = []

Read the actual TCP position

nRet = cps.HRIF_ReadActTcpPos(0,0,result)

#Read the TCP actual position variable

dX = float(result[0]) dY = float(result[1]) dZ = float(result[2])

dRx = float(result[3]) dRy = float(result[4]) dRz = float(result[5])

3.6.6.HRIF_ReadCmdJointVel

3.6.6.1.Description: Read the joint command speed.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint speed	float	-	dJ1: Joint1 command speed, unit[°/s] dJ2: Joint2 command speed, unit[°/s] dJ3: Joint3 command speed, unit[°/s] dJ4: Joint4 command speed, unit[°/s] dJ5: Joint5 command speed, unit[°/s] dJ6: Joint6 command speed, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.6.2.Example

Define the return value-empty list

result = []

Read the joint command speed

nRet = cps.HRIF_ReadCmdJointVel(0,0,result)

Read the joint command speed variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

3.6.7.HRIF_ReadActJointVel

3.6.7.1.Description: Read the actual joint speed.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint speed	float	-	dJ1: Joint1 actual speed, unit[°/s] dJ2: Joint2 actual speed, unit[°/s] dJ3: Joint3 actual speed, unit[°/s] dJ4: Joint4 actual speed, unit[°/s] dJ5: Joint5 actual speed, unit[°/s] dJ6: Joint6 actual speed, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.7.2.Example

Define the return value-empty list

result = []

Read the actual joint speed

nRet = cps.HRIF_ReadActJointVel(0,0,result)

Read the joint joint actual velocity variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

3.6.8.HRIF_ReadCmdTcpVel

3.6.8.1.Description: Read the command TCP speed.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	TCP speed	float	-	dX: X speed, unit[mm/s] dY: Y speed, unit[mm/s] dZ: Z speed, unit[mm/s] dRx: Rx speed, unit[°/s] dRy: Ry speed, unit[°/s] dRz: Rz speed, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.8.2.Example

Define the return value-empty list

result = []

Read the TCP command speed

nRet = cps.HRIF_ReadCmdTcpVel(0,0,result)

Read the TCP command speed variable

dX = float(result[0]) dY = float(result[1]) dZ = float(result[2])

dRx = float(result[3]) dRy = float(result[4]) dRz = float(result[5])

3.6.9.HRIF_ReadActTcpVel

3.6.9.1.Description: Read the actual TCP speed.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	TCP speed	float	-	dX: X speed, unit[mm/s] dY: Y speed, unit[mm/s] dZ: Z speed, unit[mm/s] dRx: Rx speed, unit[°/s] dRy: Ry speed, unit[°/s] dRz: Rz speed, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.9.2.Example

Define the return value-empty list

result = []

Read the TCP actual speed

nRet = cps.HRIF_ReadActTcpVel(0,0,result)

Read the TCP actual speed variable

dX = float(result[0]) dY = float(result[1]) dZ = float(result[2])

dRx = float(result[3]) dRy = float(result[4]) dRz = float(result[5])

3.6.10.HRIF_ReadCmdJointCur

3.6.10.1.Description: Read the joint command current.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint current	float	-	dJ1: Joint1 command current, unit[A] dJ2: Joint2 command current, unit[A] dJ3: Joint3 command current, unit[A] dJ4: Joint4 command current, unit[A] dJ5: Joint5 command current, unit[A] dJ6: Joint6 command current, unit[A]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.10.2.Example

Define the return value-empty list

result = []

Read the joint command current

nRet = cps.HRIF_ReadCmdJointCur(0,0,result)

Read the joint command current variable

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

3.6.11.HRIF_ReadActJointCur

3.6.11.1.Description: Read the actual joint current.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint current	float	-	dJ1: Joint1 actual current, unit[A] dJ2: Joint2 actual current, unit[A] dJ3: Joint3 actual current, unit[A] dJ4: Joint4 actual current, unit[A] dJ5: Joint5 actual current, unit[A] dJ6: Joint6 actual current, unit[A]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.11.2.Example

Define the return value-empty list

result = []

Read the actual joint current

nRet = cps.HRIF_ReadActJointCur(0,0,result)

Read the joint actual current variables

dJ1 = float(result[0]) dJ2 = float(result[1]) dJ3 = float(result[2])

dJ4 = float(result[3]) dJ5 = float(result[4]) dJ6 = float(result[5])

3.6.12.HRIF_ReadTcpVelocity

3.6.12.1.Description: Read the TCP end speed.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Command speed	float	-	Command speed, unit[mm/s]
result[1]	Actual velocity	float	-	Actual velocity , unit[mm/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.6.12.2.Example

Define the return value-empty list

result = []

Read the current terminal TCP speed

nRet = cps.HRIF_ReadTcpVelocity(0,0,result)

3.7.Coordinate conversion calculation instruction

3.7.1.HRIF_Quaternion2RPY

3.7.1.1.Description: Four-element turn to Euler angle.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dQuaW	W	float	-	W
dQuaX	Xi	float	-	Xi
dQuaY	Yj	float	-	Yj
dQuaZ	Zk	float	-	Zk

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Eulerian angle Rx	float	$\geq -180, < 180$	Eulerian angle Rx
result[1]	Eulerian angle Ry	float	$\geq -180, < 180$	Eulerian angle Ry
result[2]	Eulerian angle Rz	float	$\geq -180, < 180$	Eulerian angle Rz

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.1.2.Example

Define the return value-empty list

result = []

Four-element variable that needs to be converted

dQuaW = 0 dQuaX = 0 dQuaY = 0 dQuaZ = 0

Change

nRet = cps.HRIF_Quaternion2RPY(0,0,dQuaW, dQuaX, dQuaY, dQuaZ, result)

Tranconverted Euler angle results

$dRx = \text{float}(\text{result}[0])$ $dRy = \text{float}(\text{result}[1])$ $dRz = \text{float}(\text{result}[2])$

3.7.2.HRIF_RPY2Quaternion

3.7.2.1.Description: The Euler angle turns the four elements.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
Rx	Eulerian angle Rx	float	$\geq -180, \leq 180$	Eulerian angle Rx
Ry	Eulerian angle Ry	float	$\geq -180, \leq 180$	Eulerian angle Ry
Rz	Eulerian angle Rz	float	$\geq -180, \leq 180$	Eulerian angle Rz

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	W	float	-	W
result[1]	Xi	float	-	Xi
result[2]	Yj	float	-	Yj
result[3]	Zk	float	-	Zk

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.2.2.Example

Define the return value-empty list

result = []

The converted Euler corner variable that needs to be converted

dRx = 0 dRy = 0 dRz = 0

Change

nRet = cps.HRIF_RPY2Quaternion (0,0,dRx, dRy, dRz, result)

The converted four-element variable

dQuaW = float(result[0]) dQuaX = float(result[1])

dQuaY = float(result[2]) dQuaZ = float(result[3])

3.7.3.HRIF_GetInverseKin

3.7.3.1.Description: Kinematic inverse solution, calculated from the specified user coordinate system position and the dikal coordinate under the tool coordinate system.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dCoord_X- dCoord_Rz	The target Decal position of the inverse solution needs to be calculated	float	-	Target Decal position required to calculate the inverse solution: dCoord_X: X coordinate, unit[mm] dCoord_Y: Y coordinate, unit[mm] dCoord_Z: Z coordinate, unit[mm] dCoord_Rx: Rx coordinate, unit[°] dCoord_Ry: Ry coordinate, unit[°] dCoord_Rz: Rz coordinate, unit[°]
dTcp_X-dTcp_R z	Tool coordinates	float	-	Does the target location contains tool coordinate (without tool coordinate all values =0): dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dUcs_X-dUcs_R z	User coordinates	float	-	Does the target location contains user coordinate (without user coordinate all values =0): dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°]

				dUcs_Rz: Rz coordinate, unit[°]
dJ1-dJ6	Reference joint coordinates. When the inverse solution has multiple solutions, you should select the final solution according to the reference joint coordinates	float	-	dJ1: Joint1 coordinate, unit[°] dJ2: Joint2 coordinate, unit[°] dJ3: Joint3 coordinate, unit[°] dJ4: Joint4 coordinate, unit[°] dJ5: Joint5 coordinate, unit[°] dJ6: Joint6 coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Joint coordinate	float	-	dTargetJ1: Joint1 coordinate, unit[°] dTargetJ2: Joint2 coordinate, unit[°] dTargetJ3: Joint3 coordinate, unit[°] dTargetJ4: Joint4 coordinate, unit[°] dTargetJ5: Joint5 coordinate, unit[°] dTargetJ6: Joint6 coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.3.2.Example

Define the return value-empty list

result = []

Defines the spatial location variable that needs to be transformed

dCoord = [0, 0, 0, 0, 0, 0]

Define the tool coordinate variable

Tcp = [0, 0, 0, 0, 0, 0]

#Define the tool coordinate variable

Ucs = [0, 0, 0, 0, 0, 0]

Define the reference joint position variable

rawACS = [0, 0, 0, 0, 0, 0]

To reverse the solution

nRet = cps.HRIF_GetInverseKin(0,0,dCoord, rawACS, Tcp, Ucs, result)

Read the conversion results

dTargetJ1 = float(result[0]) dTargetJ2 = float(result[1]) dTargetJ3 = float(result[2])

dTargetJ4 = float(result[3]) dTargetJ5 = float(result[4]) dTargetJ6 = float(result[5])

3.7.4.HRIF_GetForwardKin

3.7.4.1.Description: A kinematic positive solution that calculates the joint coordinate position between the user coordinate system and the dikal coordinate position under the tool coordinate system.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dJ1-dJ6	The joint coordinates of the positive solution need to be calculated	float	-	dJ1: Joint 1 coordinate, unit[°] dJ2: Joint 2 coordinate, unit[°] dJ3: Joint 3 coordinate, unit[°] dJ4: Joint 4 coordinate, unit[°] dJ5: Joint 5 coordinate, unit[°] dJ6: Joint 6 coordinate, unit[°]
dTcp_X-dTcp_R z	Tool coordinate	float	-	Whether the target location contains the tool coordinate(All values without tool coordinate =0): dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dUcs_X-dUcs_R z	User coordinate	float	-	Does the target location contains user coordinate (without user coordinate all values =0): dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°] dUcs_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]]	Target Decal coordinate	float	-	If the Dear coordinate does not require tools coordinate and user coordinate, set all the tools coordinate and user coordinate 0:dTargetX: X coordinate, unit[mm] dTargetY: Y coordinate, unit[mm] dTargetZ: Z coordinate, unit[mm] dTargetRx: Rx coordinate, unit[°] dTargetRy: Ry coordinate, unit[°] dTargetRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.4.2.Example

Define the return value-empty list

result = []

Define the joint position variable that needs to be converted

rawACS = [0, 0, 0, 0, 0, 0, 0]

Define the tool coordinate variable

Tcp = [0, 0, 0, 0, 0, 0]

Define the user coordinate variable

Ucs = [0, 0, 0, 0, 0, 0]

Seek positive solution

nRet = cps.HRIF_GetForwardKin(0,0,rawACS, Tcp, Ucs, result)

Get the converted spatial position results

dTarget_X = float(result[0]) dTarget_Y = float(result[1]) dTarget_Z = float(result[2])

dTarget_Rx = float(result[3]) dTarget_Ry = float(result[4]) dTarget_Rz = float(result[5])

3.7.5.HRIF_Base2UcsTcp

3.7.5.1.Description: Calculate the location of the dikal coordinate system of the specified user and the tool coordinate system.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dCoord_X- dCoord_Rz	Deckal positions based on the pedestal coordinate system	float	-	Dekal Location to be converted: dCoord_X: X coordinate, unit[mm] dCoord_Y: Y coordinate, unit[mm] dCoord_Z: Z coordinate, unit[mm] dCoord_Rx: Rx coordinate, unit[°] dCoord_Ry: Ry coordinate, unit[°] dCoord_Rz: Rz coordinate, unit[°]
dTcp_X-dTcp_R z	Tool coordinates	float	-	Whether the target location contains tool coordinates (without all values =0): dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dUcs_X-dUcs_R z	User coordinates	float	-	Whether the target location contains user coordinates (without user coordinates, all values =0): dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°]

				dUcs_Rz: Rz coordinate, unit[°]
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✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Target Decal coordinate	float	-	If the Dear coordinate does not require tools coordinate and user coordinate, set all the tools coordinate and user coordinate 0: dTargetX: X coordinate, unit[mm] dTargetY: Y coordinate, unit[mm] dTargetZ: Z coordinate, unit[mm] dTargetRx: Rx coordinate, unit[°] dTargetRy: Ry coordinate, unit[°] dTargetRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.5.2.Example

Define the return value-empty list

result = []

Defines the spatial location variable that needs to be transformed

dCoord = [0, 0, 0, 0, 0, 0]

Define the tool coordinate variable

Tcp = [0, 0, 0, 0, 0, 0]

Define the user coordinate variable

Ucs = [0, 0, 0, 0, 0, 0]

The base coordinates are converted to the user coordinates

nRet = cps.HRIF_Base2UcsTcp(0,0,dCoord, Tcp, Ucs, result)

Define the transformed spatial location results

dTarget_X = float(result[0]) dTarget_Y = float(result[1]) dTarget_Z = float(result[2])

dTarget_Rx = float(result[3]) dTarget_Ry = float(result[4]) dTarget_Rz = float(result[5])

3.7.6.HRIF_UcsTcp2Base

3.7.6.1.Description: The coordinate position under the base coordinate system is calculated by assigning the location of the user coordinate system and the tool coordinate system.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dCoord_X- dCoord_Rz	Deckal positions based on the pedestal coordinate system	float	-	Dekal Location to be converted: dCoord_X: X coordinate, unit[mm] dCoord_Y: Y coordinate, unit[mm] dCoord_Z: Z coordinate, unit[mm] dCoord_Rx: Rx coordinate, unit[°] dCoord_Ry: Ry coordinate, unit[°] dCoord_Rz: Rz coordinate, unit[°]
dTcp_X-dTcp_Rz	Tool coordinates	float	-	Whether the target location contains tool coordinates (without all values =0): dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dUcs_X-dUcs_Rz	User coordinates	float	-	Whether the target location contains user coordinates (without user coordinates, all values =0): dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°] dUcs_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Target Decal coordinate	float	-	If the Dear coordinate does not require tools coordinate and user coordinate, set all the tools coordinate and user coordinate 0:dTargetX: X coordinate, unit[mm] dTargetY: Y coordinate, unit[mm] dTargetZ: Z coordinate, unit[mm] dTargetRx: Rx coordinate, unit[°] dTargetRy: Ry coordinate, unit[°] dTargetRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.6.2.Example

Define the return value-empty list

result = []

Defines the spatial location variable that needs to be transformed

dCoord = [0, 0, 0, 0, 0, 0]

Define the tool coordinate variable

Tcp = [0, 0, 0, 0, 0, 0]

Define the user coordinate variable

Ucs = [0, 0, 0, 0, 0, 0]

User coordinates are converted to the base coordinates

nRet = cps.HRIF_UcsTcp2Base(0,dCoord, Tcp, Ucs, result)

Read the transformed spatial position results

dTarget_X = float(result[0]) dTarget_Y = float(result[1]) dTarget_Z = float(result[2])

dTarget_Rx = float(result[3]) dTarget_Ry = float(result[4]) dTarget_Rz = float(result[5])

3.7.7.HRIF_PoseAdd

3.7.7.1.Description: point addition calculation, using the matrix left multiplication operation (the second point left by the first point).

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Rz	Spatial coordinates 1	float	-	The spatial coordinates that need to be calculated1 : dPose1_X: X coordinate, unit[mm] dPose1_Y: Y coordinate, unit[mm] dPose1_Z: Z coordinate, unit[mm] dPose1_Rx: Rx coordinate, unit[°] dPose1_Ry: Ry coordinate, unit[°] dPose1_Rz: Rz coordinate, unit[°]
dPose2_X- dPose2_Rz	Spatial coordinates 2	float	-	The spatial coordinates required to be calculated 2: dPose2_X: X coordinate, unit[mm] dPose2_Y: Y coordinate, unit[mm] dPose2_Z: Z coordinate, unit[mm] dPose2_Rx: Rx coordinate, unit[°] dPose2_Ry: Ry coordinate, unit[°] dPose2_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Computing result	float	-	Computing result: dPose3_X: X coordinate, unit[mm] dPose3_Y: Y coordinate, unit[mm] dPose3_Z: Z coordinate, unit[mm] dPose3_Rx: Rx coordinate, unit[°]

				dPose3_Ry: Ry coordinate, unit[°] dPose3_Rz: Rz coordinate, unit[°]
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✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.7.2.Example

Define the return value-empty list

result = []

Define the spatial coordinates that need to be calculated1

Pose1 = [420, 0, 445, 180, 0, 180]

Define the spatial coordinates that need to be calculated2

Pose2 = [420, 50, 445, 180, 0, 180]

Computing result

nRet = cps.HRIF_PoseAdd(0,Pose1,Pose2, result)

Computing result

dPose3_X = float(result[0]) dPose3_Y = float(result[1]) dPose3_Z = float(result[2])

dPose3_Rx = float(result[3]) dPose3_Ry = float(result[4]) dPose3_Rz = float(result[5])

3.7.8.HRIF_PoseSub

3.7.8.1.Description: the point subtraction calculation, take the second point as the reference point.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rblID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Rz	Spatial coordinates 1	float	-	The spatial coordinates that need to be calculated1 : dPose1_X: X coordinate, unit[mm] dPose1_Y: Y coordinate, unit[mm] dPose1_Z: Z coordinate, unit[mm] dPose1_Rx: Rx coordinate, unit[°] dPose1_Ry: Ry coordinate, unit[°] dPose1_Rz: Rz coordinate, unit[°]
dPose2_X- dPose2_Rz	Spatial coordinates 2	float	-	The spatial coordinates required to be calculated 2: dPose2_X: X coordinate, unit[mm] dPose2_Y: Y coordinate, unit[mm] dPose2_Z: Z coordinate, unit[mm] dPose2_Rx: Rx coordinate, unit[°] dPose2_Ry: Ry coordinate, unit[°] dPose2_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Computing result	float	-	Computing result: dPose3_X: X coordinate, unit[mm] dPose3_Y: Y coordinate, unit[mm] dPose3_Z: Z coordinate, unit[mm] dPose3_Rx: Rx coordinate, unit[°]

				dPose3_Ry: Ry coordinate, unit[°] dPose3_Rz: Rz coordinate, unit[°]
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✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.8.2.Example

Define the return value-empty list

result = []

Define the spatial coordinates that need to be calculated1

Pose1 = [420, 0, 445, 180, 0, 180]

Define the spatial coordinates that need to be calculated2

Pose2 = [420, 50, 445, 180, 0, 180]

Computing result

nRet = cps.HRIF_PoseSub(0,0,Pose1, Pose2, result)

Computing result

dPose3_X = float(result[0]) dPose3_Y = float(result[1]) dPose3_Z = float(result[2])

dPose3_Rx = float(result[3]) dPose3_Ry = float(result[4]) dPose3_Rz = float(result[5])

3.7.9.HRIF_PoseTrans

3.7.9.1.Description: Coordinate transformation.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Rz	Coordinate position 1	float	-	Coordinate position 1: dPose1_X: X coordinate, unit[mm] dPose1_Y: Y coordinate, unit[mm] dPose1_Z: Z coordinate, unit[mm] dPose1_Rx: Rx coordinate, unit[°] dPose1_Ry: Ry coordinate, unit[°] dPose1_Rz: Rz coordinate, unit[°]
dPose2_X- dPose2_Rz	Coordinate position 2	float	-	Coordinate position 2:dPose2_X: X coordinate, unit[mm] dPose2_Y: Y coordinate, unit[mm] dPose2_Z: Z coordinate, unit[mm] dPose2_Rx: Rx coordinate, unit[°] dPose2_Ry: Ry coordinate, unit[°] dPose2_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	计算 coordinate	float	-	Computing result: dPose3_X: X coordinate, unit[mm] dPose3_Y: Y coordinate, unit[mm] dPose3_Z: Z coordinate, unit[mm] dPose3_Rx: Rx coordinate, unit[°] dPose3_Ry: Ry coordinate, unit[°] dPose3_Rz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.9.2.Example

Define the return value-empty list

result = []

Define the spatial coordinates 1 that need to be calculated

Pose1 = [420, 0, 445, 180, 0, 180]

Define the spatial coordinates 2 that need to be calculated

Pose2 = [420, 50, 445, 180, 0, 180]

Computing result

nRet = cps.HRIF_PoseTrans(0,0,Pose1, Pose2, result)

Computing result

dPose3_X = float(result[0]) dPose3_Y = float(result[1]) dPose3_Z = float(result[2])

dPose3_Rx = float(result[3]) dPose3_Ry = float(result[4]) dPose3_Rz = float(result[5])

3.7.10.HRIF_PoseInverse

3.7.10.1.Description: Inverse coordinate transformation.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Rz	Space coordinates 1	float	-	The spatial coordinates 1 that need to be calculated: dPose1_X: X coordinate, unit[mm] dPose1_Y: Y coordinate, unit[mm] dPose1_Z: Z coordinate, unit[mm] dPose1_Rx: Rx coordinate, unit[°] dPose1_Ry: Ry coordinate, unit[°] dPose1_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Calculate coordinates	float	-	Computing result: dPose3_X: X coordinate, unit[mm] dPose3_Y: Y coordinate, unit[mm] dPose3_Z: Z coordinate, unit[mm] dPose3_Rx: Rx coordinate, unit[°] dPose3_Ry: Ry coordinate, unit[°] dPose3_Rz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.10.2.Example

Define the return value-empty list

```
result = [ ]  
# Define the spatial coordinates l that need to be calculated  
Pose = [420, 0, 445, 180, 0, 180]  
# Computing result  
nRet = cps.HRIF_PoseInverse(0,0,Pose1, result)  
# Computing result  
dPose3_X = float(result[0]) dPose3_Y = float(result[1]) dPose3_Z = float(result[2])  
dPose3_Rx = float(result[3]) dPose3_Ry = float(result[4]) dPose3_Rz = float(result[5])
```

3.7.11.HRIF_PoseDist

3.7.11.1.Description: Calculate the point location distance.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rblID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Rz	Space coordinates 1	float	-	Need ted spatial coordinates 1 : dPose1_X: X coordinate, unit[mm] dPose1_Y: Y coordinate, unit[mm] dPose1_Z: Z coordinate, unit[mm] dPose1_Rx: Rx coordinate, unit[°] dPose1_Ry: Ry coordinate, unit[°] dPose1_Rz: Rz coordinate, unit[°]
dPose2_X- dPose2_Rz	Space coordinates 2	float	-	The spatial coordinates required to be calculated 2: dPose2_X: X coordinate, unit[mm] dPose2_Y: Y coordinate, unit[mm] dPose2_Z: Z coordinate, unit[mm] dPose2_Rx: Rx coordinate, unit[°] dPose2_Ry: Ry coordinate, unit[°] dPose2_Rz: Rz coordinate, unit[°]

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Point distance	float	-	Point distance,unit[mm]
result[1]	Attitude distance	float	-	Attitude distance, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call

				failure
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3.7.11.2.Example

Define the return value-empty list

result = []

Define the spatial coordinates of 1 that need to be calculated

Pose1 = [420, 0, 445, 180, 0, 180]

Define the spatial coordinates of 2 that need to be calculated

Pose2 = [420, 50, 445, 180, 0, 180]

Computing result

nRet = cps.HRIF_PoseDist(0,0,Pose1, Pose2, result)

3.7.12.HRIF_Poseinterpolate

3.7.12.1.Description: Space position straight line insertion calculation.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rblID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Rz	Space coordinates 1	float	-	Need ted spatial coordinates 1 : dPose1_X: X coordinate, unit[mm] dPose1_Y: Y coordinate, unit[mm] dPose1_Z: Z coordinate, unit[mm] dPose1_Rx: Rx coordinate, unit[°] dPose1_Ry: Ry coordinate, unit[°] dPose1_Rz: Rz coordinate, unit[°]
dPose2_X- dPose2_Rz	Space coordinates 2	float	-	The spatial coordinates required to be calculated 2 : dPose2_X: X coordinate, unit[mm] dPose2_Y: Y coordinate, unit[mm] dPose2_Z: Z coordinate, unit[mm] dPose2_Rx: Rx coordinate, unit[°] dPose2_Ry: Ry coordinate, unit[°] dPose2_Rz: Rz coordinate, unit[°]
dAlpha	Addition ratio	float	0-1	dAlpha=0: dPose3=dPose1 dAlpha=1: dPose3=dPose2 0-1: Return to dPose3 at the position of dAlpha from dPose1 to dPose2

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Calculate coordinates	float	-	Computing result : dPose3_X: X coordinate, unit[mm] dPose3_Y: Y coordinate, unit[mm]

				dPose3_Z: Z coordinate, unit[mm] dPose3_Rx: Rx coordinate, unit[°] dPose3_Ry: Ry coordinate, unit[°] dPose3_Rz: Rz coordinate, unit[°]
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✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.12.2.Example

Define the return value-empty list

result = []

Define the spatial coordinates of 1 that need to be calculated

Pose1 = [420, 0, 445, 180, 0, 180]

Define the spatial coordinates of 2 that need to be calculated

Pose2 = [420, 50, 445, 180, 0, 180]

Addition ratio

dAlpha = 0.5

Computing result

nRet = cps.HRIF_Poseinterpolate(0,0,Pose1, Pose2, dAlpha, result)

Computing result

dPose3_X = float(result[0]) dPose3_Y = float(result[1]) dPose3_Z = float(result[2])

dPose3_Rx = float(result[3]) dPose3_Ry = float(result[4]) dPose3_Rz = float(result[5])

3.7.13.HRIF_PoseDefdFrame

3.7.13.1.Description: From the central rotation of the track, p1, p2 and p3 are the feature points of the selected trajectory before rotation, p4, p5 and p6 are the feature points of the selected trajectory after rotation, and Computing result is the user coordinate system of the rotation feature.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
dPose1_X- dPose1_Z	Space coordinates1	float	-	Need ted spatial coordinates 1 : dPose1_X: X coordinates, unit[mm] dPose1_Y: Y coordinates, unit[mm] dPose1_Z: Z coordinates, unit[mm]
dPose2_X- dPose2_Z	Space coordinates2	float	-	Need ted Space coordinates2 : dPose2_X: X coordinates, unit[mm] dPose2_Y: Y coordinates, unit[mm] dPose2_Z: Z coordinates, unit[mm]
dPose3_X- dPose3_Z	Space coordinates3	float	-	Need ted Space coordinates3 : dPose3_X: X coordinates, unit[mm] dPose3_Y: Y coordinates, unit[mm] dPose3_Z: Z coordinates, unit[mm]
dPose4_X- dPose4_Z	Space coordinates4	float	-	Need ted Space coordinates4 : dPose4_X: X coordinates, unit[mm] dPose4_Y: Y coordinates, unit[mm] dPose4_Z: Z coordinates, unit[mm]
dPose5_X- dPose5_Z	Space coordinates5	float	-	Need ted Space coordinates5 : dPose5_X: X coordinates, unit[mm] dPose5_Y: Y coordinates, unit[mm] dPose5_Z: Z coordinates, unit[mm]
dPose6_X-	Space	float	-	Need ted Space coordinates6 :

dPose6_Z	coordinates6			dPose6_X: X coordinates, unit[mm] dPose6_Y: Y coordinates, unit[mm] dPose2_Z: Z coordinates, unit[mm]
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✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Computing result	float	-	Computing result: dUcs_X: X coordinates, unit[mm] dUcs_Y: Y coordinates, unit[mm] dUcs_Z: Z coordinates, unit[mm] dUcs_Rx: Rx coordinates, unit[°] dUcs_Ry: Ry coordinates, unit[°] dUcs_Rz: Rz coordinates, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.7.13.2.Example

Define the return value-empty list

result = []

Define the spatial coordinates of 1 that need to be calculated

Pose1 = [0, 0, 0]

Define the spatial coordinates of 2 that need to be calculated

Pose2 = [0, 0, 0]

Define the spatial coordinates 3 that need to be calculated

Pose3 = [0, 0, 0]

Define the spatial coordinates 4 that need to be computed

Pose4 = [0, 0, 0]

Define the spatial coordinates 5 that need to be computed

Pose5 = [0, 0, 0]

Define the spatial coordinates 6 that need to be calculated

Pose6 = [0, 0, 0]

Computing result

nRet = cps.HRIF_PoseDefdFrame(0,0,Pose1, Pose2, Pose3, Pose4, Pose5, Pose6, result)

3.8. Tool coordinates and user coordinates to read and write instructions

3.8.1.HRIF_SetTCP

3.8.1.1.Description: Set the current tool coordinates, do not write to the configuration file, and become invalid after restart.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dTcp_X-dTcp_Rz	Tool coordinates	float	-	Tool coordinates that need to be set up: dTcp_X: X coordinates, unit[mm] dTcp_Y: Y coordinates, unit[mm] dTcp_Z: Z coordinates, unit[mm] dTcp_Rx: Rx coordinates, unit[°] dTcp_Ry: Ry coordinates, unit[°] dTcp_Rz: Rz coordinates, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.1.2.Example

Define the tool coordinate variable

Tcp = [0, 0, 0, 0, 0, 0]

Set tool coordinates

nRet = cps.HRIF_SetTCP(0,0,Tcp)

3.8.2.HRIF_SetUCS

3.8.2.1.Description: Set the current user coordinates, do not write to the configuration file, and become invalid after restart.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dUcs_X-dUcs_Rz	user coordinates	float	-	User coordinates that needs to be set: dUcs_X: X coordinates, unit[mm] dUcs_Y: Y coordinates, unit[mm] dUcs_Z: Z coordinates, unit[mm] dUcs_Rx: Rx coordinates, unit[°] dUcs_Ry: Ry coordinates, unit[°] dUcs_Rz: Rz coordinates, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.2.2.Example

Define the user coordinate variable

Ucs = [0, 0, 0, 0, 0, 0]

Set user coordinates

nRet = cps.HRIF_SetUCS(0,0,Ucs)

3.8.3.HRIF_ReadCurTCP

3.8.3.1.Description: Read the tool coordinate values for the current settings.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Tool coordinates	float	-	Read to the tool coordinates: dTcp_X: Xcoordinate, unit[mm] dTcp_Y: Ycoordinate, unit[mm] dTcp_Z: Zcoordinate, unit[mm] dTcp_Rx: Rxcoordinate, unit[°] dTcp_Ry: Rycoordinate, unit[°] dTcp_Rz: Rzcoordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.3.2.Example

Define the return value-empty list

result = []

Read tool coordinates

nRet = cps.HRIF_ReadCurTCP(0,0,result)

Read the tool coordinate variable

dTcp_X = float(result[0]) dTcp_Y = float(result[1]) dTcp_Z = float(result[2])

dTcp_Rx = float(result[3]) dTcp_Ry = float(result[4]) dTcp_Rz = float(result[5])

3.8.4.HRIF_ReadCurUCS

3.8.4.1.Description: Read the user coordinate value for the current setting.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	User coordinates	float	-	Read the user coordinates: dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°] dUcs_Rz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.4.2.Example

Define the return value-empty list

result = []

Read User coordinates

nRet = cps.HRIF_ReadCurUCS(0,0, result)

Read the User coordinates variable

dUcs_X = float(result[0]) dUcs_Y = float(result[1]) dUcs_Z = float(result[2])

dUcs_Rx = float(result[3]) dUcs_Ry = float(result[4]) dUcs_Rz = float(result[5])

3.8.5.HRIF_SetTCPByName

3.8.5.1.Description: Set the value in the tool coordinate list as the current tool coordinate, corresponding to the tool name of the TCP instruction of the Name for the user configuration page.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTcpName	Tool coordinates name	string	-	Name of the tool coordinates required to be set

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.5.2.Example

Name of the tool coordinate required to be issued

sTcpName = "TCP"

Set tool coordinates

nRet = cps.HRIF_SetTCPByName(0,0,sTcpName)

3.8.6.HRIF_SetUCSByName

3.8.6.1.Description: Set the value in the User coordinates list as the current User coordinates, and the corresponding name is the name of the presentation configuration page User coordinates.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sUcsName	User coordinates name	string	-	The User coordinates name that needs to be set up

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.6.2.Example

The User coordinates name that needs to be issued

sUcsName = "UCS"

Set up the User coordinates

nRet = cps.HRIF_SetUCSByName(0,0,sUcsName)

3.8.7.HRIF_ReadTCPByName

3.8.7.1.Description: Specify TCP coordinates through name reads, corresponding to the tool name for the TCP teaching device configuration page.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
sTcpName	Tool coordinates name	string	-	Name of the tool coordinate required to be read

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Tool coordinates	float	-	Read to the tool coordinates: dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.7.2.Example

Define the return value-empty list

result = []

Name of the tool coordinate required to be read

sTcpName = "TCP"

Read tool coordinates

nRet = cps.HRIF_ReadTCPByName(0,0, sTcpName,result)

Read the tool coordinate result variable

dTcp_X = float(result[0]) dTcp_Y = float(result[1]) dTcp_Z = float(result[2])

dTcp_Rx = float(result[3]) dTcp_Ry = float(result[4]) dTcp_Rz = float(result[5])

3.8.8.HRIF_ReadUCSByName

3.8.8.1.Description:Specify the UCS coordinates through the name read, and the corresponding name is the User coordinates name of the user configuration page User coordinates instruction.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
sUcsName	User coordinates Name	string	-	The User coordinates name that needs to be read

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	User coordinates	float	-	Read to the User coordinates : dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°] dUcs_Rz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.8.8.2.Example

Define the return value-empty list

result = []

The User coordinates name that needs to be read

sUcsName = "UCS"

Read User coordinates

nRet = cps.HRIF_ReadUCSByName(0,0,sUcsName , result)

Read the User coordinates variable

$dUcs_X = 0 \quad dUcs_Y = 0 \quad dUcs_Z = 0$

$dUcs_Rx = 0 \quad dUcs_Ry = 0 \quad dUcs_Rz = 0$

3.9.Force control command

3.9.1.HRIF_SetForceControlState

3.9.1.1.Description: Set the force control state, and the robot jumps to the motion state after executing the command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nState	Force control state	int	0/1-	Force control status: 0: Close the force control 1: Open the force control

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.1.2.Example

Define the setting state

nState = 1

Set the force control state

nRet = cps.HRIF_SetForceControlState(0,0,nState)

3.9.2.HRIF_ReadForceControlState

3.9.2.1.Description: Read the current force control state.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Force control state	int	0~3	Force control status: 0: Shut off status 1: Open force control exploration state 2: Force control of the complete state of exploration 3: Force control free drive state

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.2.2.Example

Define the return value-empty list

result = []

Read the force control state

nRet = cps.HRIF_ReadForceControlState(0,0,result)

Read to the force control state

nState = int(result[0])

3.9.3.HRIF_SetForceToolCoordinateMotion

3.9.3.1.Description: Set the direction of the force control coordinate system to the Tool coordinate direction mode.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nMode	Model	int	0/1	Set the force control coordinate system to the Tool direction: 0: Close 1: Open

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.3.2.Example

Define the setting state

nState = 1

Set the state of the force-control coordinate system

nRet = cps.HRIF_SetForceToolCoordinateMotion(0,0,nState)

3.9.4.HRIF_ForceControlinterrupt

3.9.4.1.Description: pause force control motion, only pause force control function, not pause motion and script.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.4.2.Example

Set the force control pause state

nRet = cps.HRIF_ForceControlinterrupt(0,0)

3.9.5.HRIF_ForceControlContinue

3.9.5.1.Description: Continue force control movement, only force control function, not movement and script.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.5.2.Example

Set the force control to continue the movement

nRet = cps.HRIF_ForceControlContinue(0,0)

3.9.6.HRIF_SetForceZero

3.9.6.1.Description: Force control zero, rescalate the force sensor on the basis of the original data.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.6.2.Example

Zero zero force control data

nRet = cps.HRIF_SetForceZero(0,0)

3.9.7.HRIF_SetMaxSearchVelocities

3.9.7.1.Description: Set the maximum speed of force control exploration.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMaxLinearVelocity	Linear velocity	float	>0	The maximum speed of exploration,unit[mm/s]
dMaxAngularVelocity	Attitude angular velocity	float	>0	The maximum speed of the exploration posture, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.7.2.Example

#Set the force control quest straight line speed

dMaxLinearVelocity = 100

Set the force control search attitude angular speed

dMaxAngularVelocity = 50

Set the force-control search speed

nRet = cps.HRIF_SetMaxSearchVelocities(0,0,dMaxLinearVelocity, dMaxAngularVelocity)

3.9.8.HRIF_SetControlFreedom

3.9.8.1.Description: Set the force control exploration degree of freedom.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nX-nRz	Each direction of freedom	int	0/1	Explore degrees of freedom switch for each axis: 0: Close 1: Open

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.8.2.Example

Define the force-control DOF state

dWrench = [0, 0, 0, 0, 0, 0]

Set the force control degree of freedom state

nRet = cps.HRIF_SetControlGoal (0,0,dWrench)

3.9.9.HRIF_SetForceControlStrategy

3.9.9.1.Description: Set the force control control policy.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nState	control strategy	int	0~2	Explore the degrees of freedom switch of each axis: 0: Constant force mode 1: Soft and smooth mode 2: Smooth and obstacle-crossing mode

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.9.2.Example

Defining force control strategies

nState = 1

Defining force control strategies

nRet = cps.HRIF_SetForceControlStrategy(0,0,nState)

3.9.10.HRIF_SetFreeDrivePositionAndOrientation

3.9.10.1.Description: Set the mounting position and attitude of the force sensor center relative to the flange disk.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	Decal coordinates	float	-	Force sensor mounting position and attitude relative to the flange plate: dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.10.2.Example

Define the force sensor installation position and attitude

dPCS = [0, 0, 0, 0, 0, 0]

Set the installation position and attitude of the force sensor

nRet = cps.HRIF_SetFreeDrivePositionAndOrientation(0,0,dPCS)

3.9.11.HRIF_SetPIDControlParams

3.9.11.1.Description: Set the force control search PID parameters.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dFp	PID parameter	float	-	PID parameterfP
dFi	PID parameter	float	-	PID parameterfI
dFd	PID parameter	float	-	PID parameterfD
dTp	PID parameter	float	-	PID parametertP
dTi	PID parameter	float	-	PID parametertI
dTd	PID parameter	float	-	PID parametertD

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.11.2.Example

Set the PID parameter

dFp = 1.0 dFi= 0.1 dFd = 0

dTp = 1.0 dTi = 0.1 dTd = 0

Set the PID parameter

nRet = cps.HRIF_SetPIDControlParams(0,0,dFp, dFi, dFd, dTp, dTi, dTd)

3.9.12.HRIF_SetMassParams

3.9.12.1.Description: Set the inertia control parameter.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	Usage control parameters	float	-	Usage control parameters : dX:X direction dY:Y direction dZ:Z direction dRx:Rx direction dRy:Ry direction dRz:Rz direction

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.12.2.Example

Set the inertia control parameters

Mass = [0, 0, 0, 0, 0, 0]

nRet = cps.HRIF_SetMassParams(0,0,Mass)

3.9.13.HRIF_SetDampParams

3.9.13.1.Description: Set the damping (b) control parameters.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	Damage control parameters	float	-	Damping control parameters: dX:X direction dY:Y direction dZ:Z direction dRx:Rx direction dRy:Ry direction dRz:Rz direction

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.13.2.Example

Set damping parameters

Damp = [800, 800, 800, 40, 40, 40]

Set damping parameters

nRet = cps.HRIF_SetDampParams(0,0,Damp)

3.9.14.HRIF_SetStiffParams

3.9.14.1.Description: Set the stiffness (k) control parameter.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	The stiffness control parameters	float	-	Stiffness control parameters: dX:X direction dY:Y direction dZ:Z direction dRx:Rx direction dRy:Ry direction dRz:Rz direction

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.14.2.Example

Set stiffness parameters

Stiff = [1000, 1000, 1000, 100, 100, 100]

Set stiffness parameters

nRet = cps.HRIF_SetStiffParams(0,0,Stiff)

3.9.15.HRIF_SetForceControlGoal

3.9.15.1.Description: Set the force control target force.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	Force control target force	float	-	Force control target force: D. X: X direction, unit [N] The dY: Y direction, unit [N] D Z: Z direction, unit [N] DRx: Rx direction, unit [NM] DRy: Ry direction, unit [NM] DdRz: Rz direction, unit [NM]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.15.2.Example

Set the force-control target force

Goal = [0, 0,10, 0, 0, 0]

Set the force control target force Z direction of 10N

nRet = cps.HRIF_SetForceControlGoal(0,0,Goal)

3.9.16.HRIF_SetControlGoal

3.9.16.1.Description: Set force control target force and target distance (0,0, force control target distance is not enabled).

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dWrench_X-dWrench_Rz	Force control target force	float	-	Force control target force: dWrenchX: X direction, unit[N] dWrenchY: Y direction, unit[N] dWrenchZ: Z direction, unit[N] dWrenchRx: Rx direction, unit[NM] dWrenchRy: Ry direction, unit[NM] dWrenchRz: Rz direction, unit[NM]
dDistance_X-dDistance_Rz	Force control target distance	float	-	Force control target distance: dDistanceX: X direction, unit[N] dDistanceY: Y direction, unit[N] dDistanceZ: Z direction, unit[N] dDistanceRx: Rx direction, unit[NM] dDistanceRy: Ry direction, unit[NM] dDistanceRz: Rz direction, unit[NM]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.16.2.Example

Set the force-control target force

Wrench = [0, 0,10, 0, 0, 0]

Set the force-control target distance

Distance = [0, 0, 0, 0, 0, 0]

Set the force control target force Z direction of 10N

nRet = cps.HRIF_SetControlGoal (0,0,Wrench, Distance)

3.9.17.HRIF_SetForceDataLimit

3.9.17.1.Description: Set the force control limit range when the force sensor exceeds this range.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMax_X-dMax_Rz	Maximum force	float	-	Maximum force range: dMax_X: X direction, unit[N] dMax_Y: Y direction, unit[N] dMax_Z: Z direction, unit[N] dMax_Rx: Rx direction, unit[NM] dMax_Ry: Ry direction, unit[NM] dMax_Rz: Rz direction, unit[NM]
dMin_X-dMin_Rz	Force minimum range	float	-	Force minimum range: dMin_X: X direction, unit[N] dMin_Y: Y direction, unit[N] dMin_Z: Z direction, unit[N] dMin_Rx: Rx direction, unit[NM] dMin_Ry: Ry direction, unit[NM] dMin_Rz: Rz direction, unit[NM]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.17.2.Example

Set the maximum force

dMax = [500, 500, 500, 500, 500, 500]

Set the force minimum

dMin = [-500, -500, -500, -500, -500, -500]

Set the force sensor data limits

nRet = cps.HRIF_SetForceDataLimit(0,0, dMax, dMin)

3.9.18.HRIF_SetForceDistanceLimit

3.9.18.1.Description: Set the force-controlled deformation range.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dAllowDistance	Maximum distance is allowed	float		Allowed for a maximum deformation distance, unit[mm]
dStrengthLevel	Power term of the distance from boundary setting	float	2/3	Power term for deviation distance between location and boundary setting: 2: The resistance is proportional to the square term of the deviation from the boundary; 3: Resistance is proportional to the cubic term that deviates from the boundary

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.18.2.Example

Set the maximum distance

dAllowDistance = 10

Set the distance from the boundary setting

dStrengthLevel = 2

Set the force-controlled deformation range

nRet = cps.HRIF_SetForceDistanceLimit(0,0, dAllowDistance, dStrengthLevel)

3.9.19.HRIF_SetForceFreeDriveMode

3.9.19.1.Description: Set the on or off force control free drive mode.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
bEnable	Whether to open	bool	False/True	False: close True: open

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.19.2.Example

Set the open force control free drive

bool bEnable = True

Set the open force control free drive

nRet = cps.HRIF_SetForceFreeDriveMode(0,0,bEnable)

3.9.20.HRIF_ReadFTCabData

3.9.20.1.Description: Read the force sensor data after calibration.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rblID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]-result[5]	Force-sensor data after the calibration	float	-	dX: X coordinate,unit[N] dY: Y coordinate,unit[N] dZ: Z coordinate,unit[N] dRx:Rx coordinate,unit[NM] dRy:Ry coordinate,unit[NM] dRz:Rz coordinate, unit[NM]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.9.20.2.Example

Define the return value-empty list

result = []

Read the force sensor data

nRet = cps.HRIF_ReadFTCabData(0,0,result)

Read to the force sensor data

dX = float(result[0]) dY = float(result[1]) dZ = float(result[2])

dRx = float(result[3]) dRy = float(result[4]) dRz = float(result[5])

3.10.General motion-type control instruction

3.10.1.HRIF_MoveRelJ

3.10.1.1.Description: the relative joint movement.

✓ Input variable

parameter	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nAxis	Axis ID	int	0~5	Target axis ID of the movement, corresponding to the joint J1-J6
nDirection	Direction	int	0/1	0: Negative direction 1: Positive direction
dDistance	Movement distance	float	-	Relative movement distance

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.1.2.Example

Defining the axis ID

nAxis = 1

Defining the direction of motion

nDirection = 1

Define movement distance

nDistance = 1

Perform the relative joint motion

nRet = cps.HRIF_MoveRelJ(0,0, nAxis, nDirection, nDistance)

3.10.2.HRIF_MoveRelL

3.10.2.1.Description: the relative movement of the space.

✓ Input variable

Parameter	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nAxis	Axis ID	int	0~5	Target axis ID of the movement, corresponding to the joint J1-J6
nDirection	Direction	int	0/1	0: Negative direction 1: Positive direction
dDistance	Movement distance	float	-	Relative movement distance
nToolMotion	Motion coordinate type	int	0/1	0: Move by the currently selected User coordinates 1: Press Tool, coordinate motion

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.2.2.Example

Defining the axis ID

nAxis = 1

Defining the direction of motion

nDirection = 1

Define movement distance

nDistance= 1

Define the motion coordinate type

nToolMotion = 1

Perform the relative spatial motion

nRet = cps.HRIF_MoveRelL(0,0, nAxis, nDirection, nDistance, nToolMotion)

3.10.3.HRIF_WayPointRel

3.10.3.1.Description: The relative movement of the road points.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nType	Type of movement	int	>0	0: Joint relative movement 1: Straight-line relative movement
nPointList	Whether to use the list point position	int	>1	0: Do not use the points in the point list 1: Use the point in the point list
Pos	locus; spatial location	list	-	nPointList= 1: Point space location dPos_X: X coordinate, unit[mm] dPos_Y: Y coordinate, unit[mm] dPos_Z: Z coordinate, unit[mm] dPos_Rx: Rx coordinate, unit[°] dPos_Ry: Ry coordinate, unit[°] dPos_Rz: Rz coordinate, unit[°] nPointList= 0, All are 0
rawACT	Joint position	list	-	nPointList= 1: Point joint position dPos_J1: Joint1coordinate, unit[°] dPos_J2: Joint2coordinate, unit[°] dPos_J3: Joint3coordinate, unit[°] dPos_J4: Joint4coordinate, unit[°] dPos_J5: Joint5coordinate, unit[°] dPos_J6: Joint6coordinate, unit[°] nPointList= 0, All are 0
nrelMoveType	Relative movement type	int	-	0: Absolute value 1: Overlay value
nAxisMask	Whether the axes are moving	list	-	Whether the axis / direction moves 0: No exercise 1: Exercise
dTarget	Movement distance	float	-	NType = 0 and nAxisMask= 1: absolute distance or superposition distance in this

				direction NType = 1 and nAxisMask= 1: absolute distance or superposition distance of the axis nAxisMask= 0: Invalid
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used
dVelocity	Velocity	float	-	Maximum movement velocity, The joints are moving in [°/s], X, Y, and Z units during spatial motion are [mm/s], Rx, Ry, and Rz units are [°/s]
dAcc	Acceleration	float	-	Maximum movement velocity, The joints are moving in [°/s ²], X, Y, and Z units during spatial motion are [mm/s ²], Rx, Ry, and Rz units are [°/s][°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
nIsUseJoint	Whether to use the joint angle	int	0	The default is 0 (other values are invalid), and joint angles are used by default
nIsSeek	Check for the DI stop	int	0/1	If nIsSeek is 1, the detection DI is stopped. If the state of the state of the DI of the nIOBit bit index of the electric box = nIOState, the robot stops the movement, otherwise the movement goes to the target point to complete the movement
nIOBit	The DI index of the detection	int	0~7	Electric box corresponds to the DI index, not valid at nIsSeek= 0
nIOState	The DI status of the detected state	int	0/1	DI status detected,, not valid at nIsSeek= 0
strCmdID	Order ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer	nRet=0 : Returns that the function call was

			value > 0	successful nRet>0 : Returns the error code of the call failure
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3.10.3.2.Example

```

# Define the type of motion
nType = 0;
# Define where the point list is used
nPointList= 0;
# Defines the spatial target location
Pos = [0, 0, 0, 0, 0, 0]
# Define the joint target locations
rawACT = [0, 0, 0, 0, 0, 0]
# Define the relative movement type
nrelMoveType= 1;
# Define whether each axis is moving in each direction
nAxisMask = [1, 1, 0, 0, 0, 0]
# Define movement distance
nTarget = [10, -10, 0, 0, 0, 0]
# Define the tool coordinate variable
sTcpName = "TCP";
# Define the User coordinates variable
sUcsName = "Base";
# Define movement speed
dVelocity = 50;
# Define the movement acceleration
dAcc = 50;
# Define the transition radius
dRadius = 50;
# Define whether the joint angle is used
nIsUseJoint = 0;
# Define whether to stop using the test DI
nIsSeek = 0;
# Define the DI index for the detection
nIOBit = 0;
# Defines the DI status
nIOState = 0;
# Define the point ID
strCmdID = "0";
# Road point relative movement

```

int nRet = HRIF_WayPointRel(0,0,nType, nPointList, Pos, rawACT, nrelMoveType, nAxisMask, nTarget, sTcpName, sUcsName, dVelocity, dAcc, dRadius, nIsUseJoint, nIsSeek, nIOBit, nIOState, strCmdID)

3.10.4.HRIF_IsMotionDone

3.10.4.1.Description: Determine whether the robot is in a motion state.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Return value	string	False/True	False: Exercise is not completed and is in motion True: Exercise is complete, and it is in a readiness state

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.4.2.Example

Define the return value-empty list

result = []

Determine whether the robot is in a motion state

nRet = cps.HRIF_IsMotionDone(0,0,result)

The robot is in a motion state

result[0] = 'False'

The robot is not in a motion state

Result[0] = 'True'

3.10.5.HRIF_IsBlendingDone

3.10.5.1.Description: Determine whether the road point is complete.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rblID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

**Note: This interface only determines whether the road point is complete, and the robot may still be in motion. For example, after following, force control, relative tracking, the robot is in motion, when the road point is not called, this interface returns to True.*

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Return value	string	False/True	False: Exercise is not completed and is in motion True: Exercise is complete, and it is in a readiness state

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.5.2.Example

Define the return value-empty list

result = []

Determine whether the road point movement is completed

nRet = cps.HRIF_IsBlendingDone(0,0,result)

The road movement is not completed

result[0] = 'False'

Road point movement completed

Result[0] ='True'

3.10.6.HRIF_WayPointEx

3.10.6.1.Description: waypoint motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nMoveType	Type of movement	int	0/1	0: Joint movement 1: Space movement
dX-dRz	Spatial target location	float	-	nMoveType=0 and nIsUseJoint=1: Invalid, nMoveType=0 and nIsUseJoint=0: use this spatial coordinate as the target position, and calculate the joint coordinate as the target joint coordinate through the inverse solution nMoveType=1: Target space locationdX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
dJ1-dJ6	Joint target location	float	-	nMoveType=0 and nIsUseJoint=1: Use this joint coordinate as the target joint coordinate nMoveType=0 and nIsUseJoint=0: This joint coordinate is only the reference joint coordinate for the selected solutions when calculating the inverse solution nMoveType=1: Invalid dJ1: Joint 1 coordinate, unit[°] dJ2: Joint 2 coordinate, unit[°] dJ3: Joint 3 coordinate, unit[°] dJ4: Joint 4 coordinate, unit[°] dJ5: Joint 5 coordinate, unit[°] dJ6: Joint 6 coordinate, unit[°]
dTcp_X-dTcp_Rz	Tool coordinate value	float	-	The tool coordinate system Name corresponds to the Name of the display page and is invalid when nIsUseJoint=1 and can be set to

				0:dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dUcs_X-dUcs_Rz	User coordinates	float	-	The User coordinates system Name of the target space coordinate corresponds to the Name of the display device page, which is invalid when nIsUseJoint=1 and can be set to 0: dUcs_X: X coordinate, unit[mm] dUcs_Y: Y coordinate, unit[mm] dUcs_Z: Z coordinate, unit[mm] dUcs_Rx: Rx coordinate, unit[°] dUcs_Ry: Ry coordinate, unit[°] dUcs_Rz: Rz coordinate, unit[°]
dVelocity	Velocity	float	-	Maximum movement velocity, The joints are moving in [°/s], X, Y, and Z units during spatial motion are [mm/s], Rx, Ry, and Rz units are [°/s]
dAcc	Acceleration	float	-	Maximum movement velocity, The joints are moving in [°/s ²], X, Y, and Z units during spatial motion are [mm/s ²], Rx, Ry, and Rz units are [°/s][°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
nIsUseJoint	Whether to use the joint coordinates	int	0/1	Whether to use joint angle as target, if nMoveType=0, nIsJoint is valid: 0: No joint angle is used 1: Use the joint angle
nIsSeek	Check for the DI stop	int	0/1	If nIsSeek is 1, the detection DI is stopped. If the state of the state of the DI of the nIOBit bit index of the electric box =nIOState, the robot stops the movement, otherwise the movement goes to the target point to complete the movement

nIOBit	The DI index of the detection	int	0-7	Electric box corresponds to the DI index, not valid at nIsSeek=0
nIOState	The DI status of the detected state	int	0/1	DI status detected,, not valid at nIsSeek=0
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

* Note: HRIF _ WayPointEx needs to set the specific value of tool coordinates and User coordinates; HRIF _ WayPoint uses the tool coordinates and User coordinates Name of the presentation.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.6.2.Example

Define the type of motion

nMoveType = 0

Defines the spatial target location

Point= [500, 500, 500, 500, 500]

Define the joint target locations

RawACSpoints = [0, 0, 0, 0, 0, 0]

Define the tool coordinate variable

Tcp = [0, 0, 0, 0, 0, 0]

Define the User coordinates variable

Ucs = [0, 0, 0, 0, 0, 0]

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 50

Define the transition radius

dRadius = 50

Define whether the joint angle is used

nIsUseJoint = 1

Define whether to stop using the test DI

nIsSeek = 0

Define the DI index for the detection

nIOBit = 0

Defines the DI status

nIOState = 0

Defining Road Point ID

stdCmdID = "0"

Perform road point movement

nRet = cps.HRIF_WayPointEx(0,0,nMoveType , Point, rawACS,Tcp, Ucs, ,dVelocity, dAcc, dRadius,
nIsUseJoint, nIsSeek, nIOBit, nIOState, strCmdID)

3.10.7.HRIF_WayPoint

3.10.7.1.Description: Roadpoint motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nMoveType	Type of movement	int	0/1	0: Joint movement 1: Straight line movement
dX-dRz	Spatial target location	float	-	nMoveType=0 and nIsUseJoint=1: Invalid, nMoveType=0 and nIsUseJoint=0: use this spatial coordinate as the target position, and calculate the joint coordinate as the target joint coordinate through the inverse solution nMoveType=1: Target space location dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]
dJ1-dJ6	Joint target location	float	-	nMoveType=0 and nIsUseJoint=1: Use this joint coordinate as the target joint coordinate nMoveType=0 and nIsUseJoint=0: This joint coordinate is only the reference joint coordinate for the selected solutions when calculating the inverse solution nMoveType=1: Invalid dJ1: 关节 1 coordinate, unit[°] dJ2: 关节 2 coordinate, unit[°] dJ3: 关节 3 coordinate, unit[°] dJ4: 关节 4 coordinate, unit[°] dJ5: 关节 5 coordinate, unit[°] dJ6: 关节 6 coordinate, unit[°]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the

				name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used
dVelocity	Velocity	float	-	Maximum movement velocity, The joints are moving in [°/s], X, Y, and Z units during spatial motion are [mm/s], Rx, Ry, and Rz units are [°/s]
dAcc	Acceleration	float	-	Maximum movement velocity, The joints are moving in [°/s ²], X, Y, and Z units during spatial motion are [mm/s ²], Rx, Ry, and Rz units are [°/s][°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
nIsUseJoint	Whether to use the joint coordinates	int	0/1	Whether to use joint angle as target, if nMoveType=0, nIsJoint is valid: 0: No joint angle is used 1: Use the joint angle
nIsSeek	Check for the DI stop	int	0/1	If nIsSeek is 1, the detection DI is stopped. If the state of the state of the DI of the nIOBit bit index of the electric box =nIOState, the robot stops the movement, otherwise the movement goes to the target point to complete the movement
nIOBit	The DI index of the detection	int	0~7	Electric box corresponds to the DI index, not valid at nIsSeek=0
nIOState	The DI status of the detected state	int	0/1	DI status detected,, not valid at nIsSeek=0
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

* Note: HRIF _ WayPointEx needs to set the specific value of tool coordinates and User coordinates; HRIF _ WayPoint uses the tool coordinates and User coordinates Name of the presentation.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call

				failure
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3.10.7.2.Example

```

# Define the type of motion
nMoveType = 0
# Defines the spatial target location
Point = [0, 0, 0, 0, 0, 0]
# Define the joint target locations
rawACS = [0, 0, 0, 0, 0, 0]
# Define the tool coordinate variable
sTcpName = "TCP"
# Define the user coordinates variable
sUcsName = "Base"
# Define movement speed
dVelocity = 50
# Define the movement acceleration
dAcc = 50
# Define the transition radius
dRadius = 50
# Define whether the joint angle is used
nIsUseJoint= 1
# Define whether to stop using the test DI
nIsSeek = 0
# Define the DI index for the detection
nIOBit = 0
# Defines the DI status
nIOState = 0
# Defining Road Point ID
stdCmdID = "0"
# Perform road point movement
nRet = cps.HRIF_WayPoint(0,0,nMoveType , Point, rawACS, sTcpName , sUcsName, dVelocity, dAcc,
dRadius,
nIsUseJoint, nIsSeek, nIOBit, nIOState, strCmdID)

```

3.10.8.HRIF_WayPoint2

3.10.8.1.Description: Waypoint movement.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nMoveType	Type of movement	int	0/1	0: Joint movement 1: Straight line movement 2: Arc movement
dEndPos_X-dEndPos_R z	Spatial target location	float	-	nMoveType=0 and nIsUseJoint=1: Invalid, nMoveType=0 and nIsUseJoint=0: use this space coordinate as the target position, and calculate the joint coordinate as the target joint coordinate by the inverse solutionnMoveType=1: Target space location nMoveType=2: As the target position for the circular arc dEndPos_X: X coordinate, unit[mm] dEndPos_Y: Y coordinate, unit[mm] dEndPos_Z: Z coordinate, unit[mm] dEndPos_Rx: Rx coordinate, unit[°] dEndPos_Ry: Ry coordinate, unit[°] dEndPos_Rz: Rz coordinate, unit[°]
dAuxPos_X-dAuxPos_ Rz	Spatial target location	float	-	nMoveType=0: Invalid nMoveType=1: Invalid nMoveType=2: As the passing position of the circular arc dAuxPosX_X: X coordinate, unit[mm] dAuxPosX_Y: Y coordinate, unit[mm] dAuxPosX_Z: Z coordinate, unit[mm] dAuxPosX_Rx: Rx coordinate, unit[°] dAuxPosX_Ry: Ry coordinate, unit[°] dAuxPosX_Rz: Rz coordinate, unit[°]



dJ1-dJ6	Joint target location	float	-	nMoveType=0 and nIsUseJoint=1: Use this joint coordinate as the target joint coordinate nMoveType=0 and nIsUseJoint=0: This joint coordinate is only the reference joint coordinate for calculating the inverse solution nMoveType=1: Invalid nMoveType=2: Invalid dJ1: Joint1 coordinate, unit[°] dJ2: Joint2 coordinate, unit[°] dJ3: Joint3 coordinate, unit[°] dJ4: Joint4 coordinate, unit[°] dJ5: Joint5 coordinate, unit[°] dJ6: Joint6 coordinate, unit[°]
sTepName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used
dVelocity	Velocity	float	-	Maximum movement velocity, The joints are moving in [°/s], X, Y, and Z units during spatial motion are [mm/s], Rx, Ry, and Rz units are [°/s]
dAcc	Acceleration	float	-	Maximum movement velocity, The joints are moving in [°/s ²], X, Y, and Z units during spatial motion are [mm/s ²], Rx, Ry, and Rz units are [°/s][°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
nIsUseJoint	Whether to use a joint coordinate	int	0/1	Whether to use joint angle as target, if nMoveType=0, nIsJoint is valid: 0: No joint angle is used

				1: Use the joint angle
nIsSeek	Check for the DI stop	int	0/1	If nIsSeek is 1, the detection DI is stopped. During the Waypoint movement, if the DI of the nIOBit bit index of the electric box =nIOState, the robot stops moving, otherwise the movement goes to the target point to complete the movement
nIOBit	The DI index of the detection	int	0~7	Electric box corresponds to the DI index, not valid at nIsSeek=0
nIOState	The DI status of the detected state	int	0/1	DI status detected,, not valid at nIsSeek=0
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

*Note: HRIF _ WayPoint2 supports line-arc transition without deceleration; HRIF _ WayPoint only supports line-line transition without deceleration.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.8.2.Example

```
# Define the type of motion
nMoveType = 0
# Defines the spatial target location
EndPos = [420, 0, 445, 180, 0, 0, 180]
# Defines the spatial target location
AuxPos = [420, 0, 445, 180, 0, 0, 180]
# Define the joint target locations
AcsPose = [0, 0, 90, 0, 90, 0]
# Define the tool coordinate variable
sTcpName = "TCP"
# Define the user coordinates variable
sUcsName = "Base"
# Define movement speed
dVelocity = 50
# Define the movement acceleration
dAcc = 50
# Define the transition radius
dRadius = 50
```

Define whether the joint angle is used

nIsUseJoint= 1

Define whether to stop using the test DI

nIsSeek = 0

Define The DI index of the detection

nIOBit = 0

Define The DI status of the detected state

nIOState = 0

Defining Road Point ID

stdCmdID = "0"

Perform road point movement

nRet = cps.HRIF_WayPoint2(0,0,nMoveType ,EndPose, AuxPos, AcsPose, sTcpName , sUcsName, dVelocity,
dAcc, dRadius, nIsUseJoint, nIsSeek, nIOBit, nIOState, strCmdID)

3.10.9.HRIF_MoveJ

3.10.9.1.Description: Joint movement.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	Spatial target location	float	-	nIsUseJoint=1: Invalid nIsUseJoint=0: Use this space coordinate as the target position, and calculate the joint coordinate as the target joint coordinate by the inverse solution: dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dJ1-dJ6	Joint target location	float	-	nIsUseJoint=1: Use this joint coordinate as the target joint coordinate nIsUseJoint=0: This joint coordinate is only used as the reference joint coordinate for calculating the inverse solution: dJ1: Joint 1 coordinate, unit[°] dJ2: Joint 2 coordinate, unit[°] dJ3: Joint 3 coordinate, unit[°] dJ4: Joint 4 coordinate, unit[°] dJ5: Joint 5 coordinate, unit[°] dJ6: Joint 6 coordinate, unit[°]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used
sUcsName	User coordinates	string	-	The name of the User coordinates system where

	name			the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used
dVelocity	Velocity	float	-	Maximum movement velocity, The joints are moving in [°/s]
dAcc	Acceleration	float	-	Maximum movement acceleration, unit[°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
nIsUseJoint	Whether to use a joint coordinate	int	0/1	Whether to use joint angle as target, if nMoveType=0, nIsJoint is valid: 0: No joint angle is used 1: Use the joint angle
nIsSeek	Check for the DI stop	int	0/1	If nIsSeek is 1, the detection DI is stopped. During the Waypoint movement, if the DI of the nIOBit bit index of the electric box =nIOState, the robot stops moving, otherwise the movement goes to the target point to complete the movement
nIOBit	The DI index of the detection	int	0~7	Electric box corresponds to the DI index, not valid at nIsSeek=0
nIOState	The DI status of the detected state	int	0/1	DI status detected,, not valid at nIsSeek=0
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.9.2.Example

Defines the spatial target location

Points = [0, 0, 90, 0, 90, 0]

Define the joint target locations

RawACSpnts = [0, 0, 90, 0, 90, 0]

Define the tool coordinate variable

sTcpName = "TCP"

Define the user coordinates variable

sUcsName = "Base"

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 50

Define the transition radius

dRadius = 50

Define whether the joint angle is used

nIsUseJoint= 1

Define whether to stop using the test DI

nIsSeek = 0

Define The DI index of the detection

nIOBit = 0

Define The DI status of the detected state

nIOState = 0

Defining Road Point ID

stdCmdID = "0"

Perform road point movement

nRet = cps.HRIF_MoveJ(0,0, Point, RawACSpots, sTepName , sUcsName, dVelocity, dAcc,
dRadius,nIsUseJoint, nIsSeek, nIOBit, nIOState, strCmdID)

3.10.10.HRIF_MoveL

3.10.10.1.Description: straight trajectory movement.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX-dRz	Spatial target location	float	-	Target space location: dTcp_X: X coordinate, unit[mm] dTcp_Y: Y coordinate, unit[mm] dTcp_Z: Z coordinate, unit[mm] dTcp_Rx: Rx coordinate, unit[°] dTcp_Ry: Ry coordinate, unit[°] dTcp_Rz: Rz coordinate, unit[°]
dJ1-dJ6	Joint reference position	float	-	Joint Reference Location-Recommended joint coordinate near the target Decal coordinate: dJ1: Joint 1 coordinate, unit[°] dJ2: Joint 2 coordinate, unit[°] dJ3: Joint 3 coordinate, unit[°] dJ4: Joint 4 coordinate, unit[°] dJ5: Joint 5 coordinate, unit[°] dJ6: Joint 6 coordinate, unit[°]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used
dVelocity	Velocity	float	-	Maximum movement velocity, X, Y, Z in [mm/s], Rx, Ry, Rz in [°/s]
dAcc	Acceleration	float	-	Maximum movement acceleration, X, Y, Z in

				[mm/s ²], Rx, Ry, Rz in[°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
nIsSeek	Check for the DI stop	int	0/1	If nIsSeek is 1, the detection DI is stopped. During the Waypoint movement, if the DI of the nIOBit bit index of the electric box =nIOState, the robot stops moving, otherwise the movement goes to the target point to complete the movement
nIOBit	The DI index of the detection	int	0~7	Electric box corresponds to the DI index, not valid at nIsSeek=0
nIOState	The DI status of the detected state	int	0/1	DI status detected,, not valid at nIsSeek=0
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.10.2.Example

Defines the spatial target location

Point = [420, 0, 445, 180, 0, 180]

Define the joint target locations

RawACSpoints = [0, 0, 90, 0, 90, 0]

Define the tool coordinate variable

sTcpName = "TCP"

Define the user coordinates variable

sUcsName = "Base"

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 50

Define the transition radius

dRadius = 50

Define whether to stop using the test DI

nIsSeek = 0

Define The DI index of the detection

nIOBit = 0

Define The DI status of the detected state

nIOState = 0

Defining Road Point ID

stdCmdID = "0"

Perform road point movement

nRet = cps.HRIF_MoveL(0,0, Point, RawACSpoints sTcpName, sUcsName, dVelocity, dAcc, dRadius,nIsSeek,
nIOBit, nIOState, strCmdID)

3.10.11.HRIF_MoveC

3.10.11.1.Description: Arc trajectory motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dStartPoint	Arc starting point position	list[float]	-	Arc starting point position: dStartPos_X: X coordinate, unit[mm] dStartPos_Y: Y coordinate, unit[mm] dStartPos_Z: Z coordinate, unit[mm] dStartPos_Rx: Rx coordinate, unit[°] dStartPos_Ry: Ry coordinate, unit[°] dStartPos_Rz: Rz coordinate, unit[°]
dAuxPoint	The arc through the point position	list[float]	-	The arc through the point position: dAuxPos_X: X coordinate, unit[mm] dAuxPos_Y: Y coordinate, unit[mm] dAuxPos_Z: Z coordinate, unit[mm] dAuxPos_Rx: Rx coordinate, unit[°] dAuxPos_Ry: Ry coordinate, unit[°] dAuxPos_Rz: Rz coordinate, unit[°]
dEndPoint	Arc end point position	list[float]	-	Arc end point positio . If the whole circle is used, the junction speed point is also a passing point on the circle, and the whole circle stops after the run: dEndPos_X: X coordinate, unit[mm] dEndPos_Y: Y coordinate, unit[mm] dEndPos_Z: Z coordinate, unit[mm] dEndPos_Rx: Rx coordinate, unit[°] dEndPos_Ry: Ry coordinate, unit[°] dEndPos_Rz: Rz coordinate, unit[°]
nFixedPosure	Whether fixed posture	int	0/1	Does the posture remain unchanged during the whole arc movement:

				0: Fixed posture 1: No fixed posture
nMoveCType	Arc type	int	0/1	0: Round 1: The arc
dRadLen	Arc length	float	-	When the arc movement is invalid, the arc path is determined by three points. When the whole circle number of the whole circle movement is used, the decimal part is invalid.
dVelocity	Velocity	float	-	Maximum movement velocity, X, Y, Z in [mm/s], Rx, Ry, Rz in [°/s]
dAcc	Acceleration	float	-	Maximum movement acceleration, X, Y, Z in [mm/s ²], Rx, Ry, Rz in [°/s ²]
dRadius	Knuckle radius	float	-	Knuckle radius, unit[mm]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.11.2.Example

Arc starting point position

dStartPoint = [420, 0, 445, 180, 0, 180]

The arc through the point position

dAuxPoint = [420, 50, 445, 180, 0, 180]

Arc end point position

dEndPoint = [470, 0, 445, 180, 0, 180]

#Whether fixed posture

nFixedPosure = 0

Arc type

nMoveCType = 0

Number of circles

dRadLen = 1

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 50

Define the transition radius

dRadius = 50

Define the tool coordinate variable

sTcpName = "TCP"

Define the user coordinates variable

sUcsName = "Base"

Defining Road Point ID

stdCmdID = "0"

Perform road point movement

nRet = cps.HRIF_MoveC(0,0,dStartPoint , dAuxPoint, dEndPoint,

nFixedPosure, nMoveCType, dRadLen, dVelocity, dAcc, dRadius,sTcpName , sUcsName, strCmdID)

3.10.12.HRIF_MoveZ

✓ Description: Z-type trajectory movementInput variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dStartPos	Z-type at the position of the start point	float	-	Z-Type Start Point Location: AA-dStartPos_X: X coordinate, unit[mm] dStartPos_Y: Y coordinate, unit[mm] dStartPos_Z: Z coordinate, unit[mm] dStartPos_Rx: Rx coordinate, unit[°] dStartPos_Ry: Ry coordinate, unit[°] dStartPos_Rz: Rz coordinate, unit[°]
dEndPos	Z-type end point position	float	-	Z-type end point position: dEndPos_X: X coordinate, unit[mm] dEndPos_Y: Y coordinate, unit[mm] dEndPos_Z: Z coordinate, unit[mm] dEndPos_Rx: Rx coordinate, unit[°] dEndPos_Ry: Ry coordinate, unit[°] dEndPos_Rz: Rz coordinate, unit[°]
dPlanePos	The trajectory determines the plane point position	float	-	Position of Z-shaped trajectory: dPlanePosPos_X: X coordinate, unit[mm] dPlanePosPos_Y: Y coordinate, unit[mm] dPlanePosPos_Z: Z coordinate, unit[mm] dPlanePosPos_Rx: Rx coordinate, unit[°] dPlanePosPos_Ry: Ry coordinate, unit[°] dPlanePosPos_Rz: Rz coordinate, unit[°]
dVelocity	Velocity	float	-	Maximum movement velocity, X, Y, Z in [mm/s], Rx, Ry, Rz in [°/s]
dAcc	Acceleration	float	-	Maximum movement acceleration, X, Y, Z in [mm/s ²], Rx, Ry, Rz in [°/s ²]
dWidth	Width	float	-	Track width

dDensity	Density	float	-	Track density, density is automatically calculated by velocity when density is not used
nEnableDensity	Whether density is used	int	-	Whether Density is used: 0: No use 1: Use
nEnablePlane	Whether to use the plane points	int	-	Whether to use plane points, and if not, determine the XYZ planes against the selected User coordinates: 0: No use 1: Use
nEnableWaiTime	Whether to open the turning point waiting time	int	-	Open turning point waiting time: 0: No use 1: Use
nPosiTime	Forward turning point waiting time	int	-	Forward turning point waiting time, unit[ms]
nNegaTime	Negative turning point waiting time	int	-	Negative turning point waiting time, unit[ms]
dRadius	knuckle radius	float	-	knuckle radius , unit[mm]
sTepName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used.
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used.
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.12.1.Example

Location of the starting point

dStartPos = [420, 0, 445, 180, 0, 180]

End point position

dEndPos = [420, 100, 445, 180, 0, 180]

Determine the trajectory plane point position

dPlanePos = [470, 50, 445, 180, 0, 180]

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 2500

Width

dWidth = 50

Density

dDensity = 10

Use density

nEnableDensity = 1

Use a plane point

nEnablePlane = 1

Whether waiting at a turning point- -not waiting

nEnableWaiTime = 0

Forward turning point waiting time

nPosiTime = 0

Negative turning point waiting time

nNegaTime = 0

Define the transition radius

dRadius = 5

Define the tool coordinate variable

sTcpName = "TCP "

Define the user coordinates variable

sUcsName = "Base"

Defining Road Point ID

stdCmdID = "0"

Perform road point movement

nRet = cps.HRIF_MoveZ(0,0,dStartPos, dEndPos, dPlanePos, dVelocity, dAcc, dWidth, dDensity,
nEnableDensity, nEnablePlane, nEnableWaiTime, nPosiTime, nNegaTime, dRadius, sTcpName, sUcsName,
strCmdID)

3.10.13.HRIF_MoveE

3.10.13.1.Description: Elliptical trajectory motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dP1	Showing position 1	list[float]	-	Showing position 1: dP1_X: X coordinate, unit[mm] dP1_Y: Y coordinate, unit[mm] dP1_Z: Z coordinate, unit[mm] dP1_Rx: Rx coordinate, unit[°] dP1_Ry: Ry coordinate, unit[°] dP1_Rz: Rz coordinate, unit[°]
dP2	Showing position 2	list[float]	-	Showing position 2: dP2_X: X coordinate, unit[mm] dP2_Y: Y coordinate, unit[mm] dP2_Z: Z coordinate, unit[mm] dP2_Rx: Rx coordinate, unit[°] dP2_Ry: Ry coordinate, unit[°] dP2_Rz: Rz coordinate, unit[°]
dP3	Showing position3	list[float]	-	Showing position3: dP3_X: X coordinate, unit[mm] dP3_Y: Y coordinate, unit[mm] dP3_Z: Z coordinate, unit[mm] dP3_Rx: Rx coordinate, unit[°] dP3_Ry: Ry coordinate, unit[°] dP3_Rz: Rz coordinate, unit[°]
dP4	Showing position4	list[float]	-	Showing position4: dP4_X: X coordinate, unit[mm]

				dP4_Y: Y coordinate, unit[mm] dP4_Z: Z coordinate, unit[mm] dP4_Rx: Rx coordinate, unit[°] dP4_Ry: Ry coordinate, unit[°] dP4_Rz: Rz coordinate, unit[°]
dP5	Showing position 5	list[float]	-	Showing position5: dP5_X: X coordinate, unit[mm] dP5_Y: Y coordinate, unit[mm] dP5_Z: Z coordinate, unit[mm] dP5_Rx: Rx coordinate, unit[°] dP5_Ry: Ry coordinate, unit[°] dP5_Rz: Rz coordinate, unit[°]
nOrientMode	Motor pattern	int	0/1	0: Oval arc 1: The whole ellipse
nMoveType	Type of movement	int	0/1	0: Do not use a fixed posture 1: Use a fixed posture
dArcLength	Arc length	float	-	arc length, unit[°]
dVelocity	Velocity	float	-	Maximum movement velocity, X, Y, Z in [mm/s], Rx, Ry, Rz in [°/s]
dAcc	Acceleration	float	-	Maximum movement acceleration, X, Y, Z in [mm/s²], Rx, Ry, Rz in [°/s²]
dRadius	Knuckle radius	float	-	Knuckle radius , unit[mm]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used.
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used.
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.13.2.Example

Showing point 1

dP1[6] = [420,0,445,180,0,180]

Showing point 2

dP2[6] = [460,0,445,180,0,180]

Showing point 3

dP3[6] = [480,10,445,180,0,180]

Showing point 4

dP4[6] = [460,20,445,180,0,180]

Showing point 5

dP5[6] = [420,20,445,180,0,180]

Move pattern

nOrientMode = 0

Type of movement

nMoveType = 1

Arc length

dArcLength = 360

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 2500

Define the transition radius

dRadius = 5

Define the tool coordinate variable

sTcpName = "TCP"

Define the user coordinates variable

sUcsName = "Base"

Defining Road Point ID

stdCmdID = "0"

Execute elliptic motion

nRet = cps.HRIF_MoveE(0,0,dP1, dP2, dP3,dP4, dP5,

nOrientMode,nMoveType,dArcLength,dVelocity,dAcc,dRadius, sTcpName, sUcsName, strCmdID)

3.10.14.HRIF_MoveS

3.10.14.1.Description: Archimedes spiral motion with an initial radius of fixed 1mm.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dSpiralIncrement	Incremental radius	float	>0	Spiral motion per coil incremental radius, unit[mm]
dSpiralDiameter	End radius	float	>1	The spiral motion end radius,unit[mm]
dVelocity	Velocity	float	-	Maximum movement velocity, unit[°/s]
dAcc	Acceleration	float	-	Maximum movement acceleration, unit[°/s ²]
dRadius	Knuckle radius	float	-	knuckle radius , unit[mm]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used.
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used.
strCmdID	Command ID	string	-	The current road point ID, which can be customized or set to "1", "2", "3" in order

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.10.14.2.Example

Define incremental radius

dSpiralIncrement = 1

Define End radius

dSpiralDiameter = 5

Define the tool coordinate variable

sTcpName = "TCP"

Define the user coordinates variable

sUcsName = "Base"

Define movement speed

dVelocity = 50

Define the movement acceleration

dAcc = 50

Define the transition radius

dRadius = 50

Defining Road Point ID

stdCmdID = "0"

Perform the spiral trajectory motion

nRet= cps.HRIF_MoveS(0,0, dSpiralIncrement, dSpiralDiameter, sTcpName, sUcsName, dVelocity, dAcc,
dRadius, sCmdID)

3.11.Continuous trajectory motion class control instruction

3.11.1.HRIF_StartPushMovePathJ

3.11.1.1.Description: Initialize continuous joint movement.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name
dSpeedRatio	Track movement Velocity ratir	float	0~1	Track movementVelocity ratir
dRadius	Knuckle radius	float	>=0	knuckle radius in [mm]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.1.2.Example

Track name

sTrackName = "Path1"

Velocity ratir

dSpeedRatio = 0.5

Knuckle radius

dRadius = 2

Initialize the joint continuous trajectory motion

nRet = cps.HRIF_StartPushMovePathJ(0,0,sTrackName, dSpeedRatio, dRadius)

3.11.2.HRIF_PushMovePathJ

3.11.2.1.Description: Issue the movement trajectory points.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name
paramsJ	Joint point	list[float]	-	Target joint location: dJ1: Joint 1 position, unit[°] dJ2: Joint 2 position, unit[°] dJ3: Joint 3 position, unit[°] dJ4: Joint 4 position, unit[°] dJ5: Joint 5 position, unit[°] dJ6: Joint 6 position, unit[°]

* Note: After calling HRIF_StartPushMovePath, this function can be called several times. Generally, the number of points should be greater than 4.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.2.2.Example

Track name

sTrackName = "Path1"

Target joint position

paramsJ = [0,0,90,0,0,0]

Issue joint points

nRet = cps.HRIF_PushMovePathJ (0,0,sTrackName, paramsJ)

3.11.3.HRIF_EndPushMovePathJ

3.11.3.1.Description: The trajectory distribution is completed and the trajectory calculation is started.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.3.2.Example

Track name

sTrackName = "Path1"

After the distribution, start calculating the track

nRet = cps.HRIF_EndPushMovePathJ(0,0,sTrackName)

3.11.4.HRIF_MovePathJ

3.11.4.1.Description: The trajectory specified by the motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.4.2.Example

Track name

sTrackName = "Path1"

movement locus

nRet = cps.HRIF_MovePathJ(0,0,sTrackName)

3.11.5.HRIF_ReadMovePathJState

3.11.5.1.Description: Read the current trajectory state.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
sTrackName	Track name	string	-	Track name

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Track state	int	0~5	Current trajectory status: 0: The trajectory does not start teaching, switch to this state after calling HRIF _ StartPushMovePathJ 1: In the track instruction, call HRIF _ PushMovePathJ 2: In the trajectory calculation, switch to this state after calling HRIF _ EndPushMovePathJ 3: After the trajectory calculation is completed, after the calculation is completed, switch to this state to start the movement. 4: End the teaching 5: calculation error, need to check whether the point is reasonable, and then correct the re-teach.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.5.2.Example

Define the return value-empty list

result = []

Track name

sTrackName = "Path1"

Read the trajectory state

nRet = cps.HRIF_ReadMovePathJState(0,0,sTrackName, result)

Current trajectory state to be read to

nState = int(result[0])

3.11.6.HRIF_UpdateMovePathJName

3.11.6.1.Description: Update the specified track name.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track original name	string	-	Track original name
sTrackNewName	Updated track name	string	-	Updated track name

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.6.2.Example

Track name

sTrackName = "Path1"

Updated track name

sTrackNewName = "Path2"

Rename Trackname

nRet = cps.HRIF_UpdateMovePathJName(0,0,sTrackName, sTrackNewName)

3.11.7.HRIF_DelMovePathJ

3.11.7.1.Description: Delete the specified trace.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.7.2.Example

Track name

sTrackName = "Path1"

Remove the trajectory

nRet = cps.HRIF_DelMovePathJ(0,0,sTrackName)

3.11.8.HRIF_ReadTrackProcess

3.11.8.1.Description: Read the current trajectory movement progress.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []

* Note: This interface is valid for HRIF_MovePathL and not for HRIF_MovePathJ.

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result[0]	Track running progress	float	0~1	Current trajectory movement progress, current movement progress (0-1), > 0.999999 indicates movement completion
result[1]	Point index	int	An integer value > 0	The point to which the current trajectory is ices

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.8.2.Example

Define the return value-empty list

result = []

Read the trajectory state

nRet = cps.HRIF_ReadTrackProcess(0,0,result)

Track running progress

dProcess = float(result[0])

Point index

nIndex = int(result[1])

3.11.9.HRIF_InitMovePathL

3.11.9.1.Description: Initializing the spatial trajectory motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name, At present, the Track name with no function of spatial trajectory movement can be arbitrarily defined, and the same trajectory operation requires the same execution of HRIF_InitMovePathL, HRIF_PushMovePaths, HRIF_EMovePathL After calling HRIF_MovePathL, the movement will start directly after calculating the trajectory. The calculation time is about 2-4s, and the movement is determined according to the actual trajectory size
dVelocity	Track motion speed	float	-	Track motion speed, unit[mm/s]
dAcc	Track motion acceleration	float	-	Track motion acceleration, unit[mm/s ²]
dJerk	Track motion jerk	float	-	Track motion jerk, unit[mm/s ³]
sTcpName	Tool coordinates name	string	-	The name of the tool coordinate system where the target space coordinate corresponds to the name of the user page, is invalid when nIsUseJoint= 1, and the default name "TCP" can be used.
sUcsName	User coordinates name	string	-	The name of the User coordinates system where the target space coordinates are located corresponds to the name of the display device page and is invalid when nIsUseJoint= 1, so the default name "Base" can be used.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.9.2.Example

```
# Track name
sTrackName = "Path1"
# Define movement speed
dVelocity = 100
```


Define the movement acceleration

dAcc = 2500

Define movement acceleration

dJerk = 1000000

Define the tool coordinate variable

sTcpName = "TCP"

Define the user coordinates variable

sUcsName = "Base"

Initialize the joint continuous trajectory motion

nRet = cps.HRIF_InitMovePathL(0,0,sTrackName, dVelocity, dAcc, dJerk, sUcsName, sTcpName)

3.11.10.HRIF_PushMovePathL

3.11.10.1.Description: Issue the movement trajectory points.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name
paramPcs	Space point	list[float]	-	Target space location : dX: X coordinate, unit[mm] dY: Y coordinate, unit[mm] dZ: Z coordinate, unit[mm] dRx: Rx coordinate, unit[°] dRy: Ry coordinate, unit[°] dRz: Rz coordinate, unit[°]

*Note: After calling HRIF_InitPushMovePathL, this function can be called several times. Generally, the number of points should exceed 4.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.10.2.Example

```
# Track name
sTrackName = "Path1"

# Defines the spatial target location
paramPcs = [420, 0, 445, 180, 0, 180]

# Issue the space target points
nRet = cps.HRIF_PushMovePathL(0,0,sTrackName, paramPcs)
```

3.11.11.HRIF_PushMovePaths

3.11.11.1.Description: Issue track points in batches, and call multiple point points data at a time.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name
nMoveType	Point type	int	0/1	0: Share joint locations with HRIF _ MovePathJ 1: Issue space points and share them with HRIF _ MovePathJ
nPointsSize	Number of points	int	-	The number of points, must be consistent with the number in the sPoints
sPoints	Point data	list[float]	-	Point point data, each data separated by a comma

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.11.2.Example

```
# Track name
sTrackName = "Path1"
# Motion Type-MovePathL
nMoveType = 1
# Number of points
nPointsSize = 6
sPoints = [420,0,445,180,0,180,420,10,445,180,0,180,420,20,445,180,0,180,
420,30,445,180,0,180,420,40,445,180,0,180,420,50,445,180,0,180]
# Issue the space target points
nRet = cps.HRIF_PushMovePaths(0,0,sTrackName, nMoveType, nPointsSize, sPoints)
# Track name
sTrackName = "Path2"
# Type of movement-MovePathJ
nMoveType = 0
# Number of points
nPointsSize = 9
```

```
sPoints = [0,0,0,0,0,0,0,0,5,0,5,0,0,0,10,0,10,0,  
0,0,15,0,15,0,0,0,20,0,20,0,0,0,25,0,25,0,  
0,0,30,0,30,0,0,0,34,0,34,0,0,0,40,0,40,0]
```

Issue joint points

```
nRet = cps.HRIF_PushMovePaths(0,0,sTrackName, nMoveType, nPointsSize, sPoints)
```

3.11.12.HRIF_MovePathL

3.11.12.1.Description: Perform the spatial coordinate trajectory motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sTrackName	Track name	string	-	Track name

* Note: After calling HRIF _ MovePathL, the movement will start directly after calculating the trajectory. The calculation time is about 2-4s, which is determined according to the actual trajectory size.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.12.2.Example

Track name

sTrackName = "Path2"

Start the spatial continuous trajectory motion

nRet = cps.HRIF_MovePathL(0,0,sTrackName)

3.11.13.HRIF_SetMovePathOverride

3.11.13.1.Description: Set MovePathVelocity ratir, the set is valid in MovePath motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
MovePathOverride	Set up velocity ratir	Float	0.01~1	Velocity ratir that needs to be set(0.01~1)

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.11.13.2.Example

Velocity ratir

MovePathOverride= 0.01

#Set up MovePathVelocity ratir

nRet = cps.HRIF_SetMovePathOverride(0,0,MovePathOverride)

3.12.The Servo motion class control instruction

3.12.1.HRIF_StartServo

3.12.1.1.Description: Set the start of the robot online control (ServoJ or ServoP).

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dServoTime	Update cycle	float	>0	Fixed cycle time of updates, unit[s]
dLookaheadTime	Forward-looking time	float	>0	Forward-looking time, unit[s]

* Note: HRIF _ StartServo / HRIF _ PushServoJ / HRIF _ PushServoP is a set of interfaces, which is not shared with other Servo instructions.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.12.1.2.Example

Period

dServoTime = 0.02

Forward-looking time

dLookaheadTime = 0.2

Start the robot for online control

nRet = cps.HRIF_StartServo(0,0,dServoTime, dLookaheadTime)

3.12.2.HRIF_PushServoJ

3.12.2.1.Description: Online joint position command control, send the joint position with the fixed update time set by StartServo, and the robot will track the joint position command in real time.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dAcs	Joint point	list[float]	-	Target joint location: dJ1: Joint 1 position, unit[°] dJ2: Joint 2 position, unit[°] dJ3: Joint 3 position, unit[°] dJ4: Joint 4 position, unit[°] dJ5: Joint 5 position, unit[°] dJ6: Joint 6 position, unit[°]

* Note: HRIF _ StartServo / HRIF _ PushServoJ / HRIF _ PushServoP is a set of interfaces, which is not shared with other Servo instructions.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.12.2.2.Example

Target joint position

dAcs = [0, 0, 0, 0, 0, 0]

Online joint position command control

nRet = cps.HRIF_PushServoJ(0,0, dAcs)

3.12.3.HRIF_PushServoP

3.12.3.1.Description: Online terminal TCP position command control, send TCP position with the fixed update time set by StartServo, and the robot will track the joint position command of the target TCP position inverse operation.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dPcs	Space point	list[float]	-	Target space location: dPcs[0]: X coordinate, unit[mm] dPcs[1]: Y coordinate, unit[mm] dPcs[2]: Z coordinate, unit[mm] dPcs[3]: Rx coordinate, unit[°] dPcs[4]: Ry coordinate, unit[°] dPcs[5]: Rz coordinate, unit[°]
dTcp	Tools coordinate	list[float]	-	Tool corresponding to the target position coordinate: dTcp[0]: X coordinate, unit[mm] dTcp[1]: Y coordinate, unit[mm] dTcp[2]: Z coordinate, unit[mm] dTcp[3]: Rx coordinate, unit[°] dTcp[4]: Ry coordinate, unit[°] dTcp[5]: Rz coordinate, unit[°]
dUcs	User coordinates	list[float]	-	The user coordinates corresponding to the target position : dUcs[0]: X coordinate, unit[mm] dUcs[1]: Y coordinate, unit[mm] dUcs[2]: Z coordinate, unit[mm] dUcs[3]: Rx coordinate, unit[°] dUcs[4]: Ry coordinate, unit[°] dUcs[5]: Rz coordinate, unit[°]

* Note: HRIF _ StartServo / HRIF _ PushServoJ / HRIF _ PushServoP is a set of interfaces, which is not shared with other Servo instructions.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.12.3.2.Example

Spatial target location

dPcs = [420, 0, 445, 180, 0, 180]

Define the tool coordinate variable

dTcp = [0, 0, 0, 0, 0, 0]

Define the user coordinates variable

dUcs = [0, 0, 0, 0, 0, 0]

Online space position command control

nRet = cps.HRIF_PushServoP(0,0,dPcs, dTcp, dUcs)

3.13.Relative tracking motion class control instruction

3.13.1.HRIF_SetMoveTraceParams

3.13.1.1.Description: Set the relative tracking motion control parameters.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nState	Tracking state	int	0/1	0: Turn off the relative tracking motion 1: Turn on the relative tracking motion
dDistance	Relative tracking motion maintained relative distance	float	>0	Relative tracking motion maintained relative distance
dAwayVelocity	Relative tracked movement away from the search speed	float	>0	Relative tracked movement away from the search speed
dBackVelocity	The near-search speed of the relative tracked motion	float	>0	The near-search speed of the relative tracked motion

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.13.1.2.Example

Set the relative distance between the tracking status and the hold

nState = 1

dDistance = 100

The search speed of relatively tracked movements

dAwayVelocity = 50

dBackVelocity = 50

Set the relative tracking motion control parameters and turn on the relative tracking motion

nRet = cps.HRIF_SetMoveTraceParams(0,0,nState, dDistance,dAwayVelocity,dBackVelocity)

3.13.2.HRIF_SetMoveTraceInitParams

3.13.2.1.Description: Set the relative tracking motion initialization parameters.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dK	Sensor calculation parameters	float	-	Computational formula $y = dK * x + dB$
dB	Sensor calculation parameters	float	-	Computing the formula $y = dK * x + dB$
dMaxLimit	The laser sensor detects the maximum distance value	float	-	The laser sensor detects the maximum distance value
dMinLimit	Laser sensor detection distance minimum value	float	-	Laser sensor detection distance minimum value

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.13.2.2.Example

Sensor calculation parameters

dK= -14

Sensor calculation parameters

dB= 135

The laser sensor detects the maximum distance value

dMaxLimit = 130

Laser sensor detection distance minimum value

dMinLimit = 65

Set the tracking state initialization parameter

nRet = cps.HRIF_SetMoveTraceInitParams(0,0,dK, dB, dMaxLimit, dMinLimit)

3.13.3.HRIF_SetMoveTraceUcs

3.13.3.1.Description: Set the tracking search direction of the relative tracking motion.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
direction	Follow up the search direction	list[float]	>-180,<180	Target space location: DX: invalid, can be set to 0 DY: invalid, can be set to 0 DZ: invalid, can be set to 0 dRx: Rx direction, unit[°] dRy: Ry direction, unit[°] dRz: Rz direction, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.13.3.2.Example

Set the tracking direction

direction = [420, 0, 445, 180, 0, 180]

Set the tracking direction

nRet = cps.HRIF_SetMoveTraceUcs(0,0,direction)

3.13.4.HRIF_SetTrackingState

3.13.4.1.Description: Set up the conveyor belt tracking motion state.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nState	Tracking state	int	0/1	0: Turn off the conveyor belt to track the movement 1: Open the conveyor belt tracking movement

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.13.4.2.Example

Set the conveyor tracking on

nState = 1

Turn on the conveyor belt tracking

nRet = cps.HRIF_SetTrackingState(0,0,nState)

3.14.The position follows the motion class control command

3.14.1.HRIF_SetPoseTrackingMaxMotionLimit

3.14.1.1.Description: Set the maximum following speed for the position to follow

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dMaxLineVel	Linear maximum speed	double	>0	Position follows straight to the maximum following speed, unit[mm/s]
dMaxOriVel	The attitude maximum speed	double	>0	Position following attitude maximum following speed, unit[°/s]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.14.1.2.Example

Set the straight-line maximum speed

dMaxLineVel = 100

Set the attitude maximum-speed

dMaxOriVel = 1

transmitted information transmitter data

nRet = cps.HRIF_SetPoseTrackingMaxMotionLimit(0,0,dMaxLineVel,dMaxOriVel)

3.14.2.HRIF_SetPoseTrackingPIDParams

3.14.2.1.Description: Set the PID parameter

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dPosPID1	Position follow PID	double	-	Position follow PID, Default =5
dPosPID2	Position follow PID	double	-	Position follow PID, default =0.1
dPosPID3	Position follow PID	double	-	Position follow PID, default =0
dOriPID1	Attitude follow PID	double	-	Attitude follow PID, default =5
dOriPID2	Attitude follow PID	double	-	Attitude follow PID, default=0.1
dOriPID3	Attitude follow PID	double	-	Attitude follow PID, default=0

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.14.2.2.Example

Set the position follow PID and stance follow PID

dPosPID1 = 5

dPosPID2 = 0.1

dPosPID3 = 0

dOriPID1 = 5

dOriPID2 = 0.1

dOriPID3 = 0

Transmitter data

nRet = cps.HRIF_SetPoseTrackingPIDParams(0,0,dPosPID1,dPosPID2,dPosPID3,dOriPID1,dOriPID2,dOriPID3)

3.14.3.HRIF_SetPoseTrackingTargetPos

3.14.3.1.Description: Set the target location where the location follows.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX	Distance maintained in the X direction	double	-	Distance maintained in the X direction, unit [mm]
dY	Distance maintained in the Y direction	double	-	Distance maintained in the Y direction, unit [mm]
dZ	Distance maintained in the Z direction	double	-	Distance maintained in the Z direction, unit[mm]
dRx	Distance maintained in the Rx direction	double	-	Distance maintained in the Rx direction, unit [°]
dRy	The distance maintained in the Ry direction	double	-	The distance maintained in the Ry direction, unit [°]
dRz	Distance maintained in the Rz direction	double	-	Distance maintained in the Rz direction, unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.14.3.2.Example

Set the target location

dX = 0 dY = 0 dZ = 100 dRx = 0 dRy = 0 dRz = 0

Transmitter data

nRet = cps.HRIF_SetPoseTrackingTargetPos(0,0,dX,dY,dZ,dRx,dRy,dRz)

3.14.4.HRIF_SetPoseTrackingState

3.14.4.1.Description: Set the position following status

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nState	The position follows the state	int	0/1	0: Close position follows 1: open the position to follow

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.14.4.2.Example

Turn on the position to follow

nState = 1

Transmitter data

nRet = cps.HRIF_SetPoseTrackingState(0,0,nState)

3.14.5.HRIF_SetUpdateTrackingPose

3.14.5.1.Description: Set up to update the sensor location information in real time.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
dX	The distance in the detected X-direction	double	-	The distance in the detected X-direction, in unit[mm]
dY	The distance in the detected Y-direction	double	-	The distance in the detected Y-direction, in unit[mm]
dZ	The distance in the detected Z-direction	double	-	The distance in the detected Z-direction, in unit[mm]
dRx	The distance in the detected Rx-direction	double	-	The distance in the detected Rx-direction, in unit[°]
dRy	The distance in the detected Ry-direction	double	-	The distance in the detected Ry-direction, in unit[°]
dRz	The distance in the detected Rz-direction	double	-	The distance in the detected Rz-direction, in unit[°]

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.14.5.2.Example

Set the detected location information

dX = 0 dY = 0 dZ = 100 dRx = 0 dRy = 0 dRz = 0

transmitter data

nRet = cps.HRIF_SetUpdateTrackingPose(0,0,dX,dY,dZ,dRx,dRy,dRz)

3.15.Other instructions

3.15.1.HRIF_HRAppCmd

3.15.1.1.Description: Execute the plug-in App command.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
sCmdName	Command name	string	0/1	Command name
sParams	List of parameters	string	-	List of parameters
result	Return value	list		Pass into the empty list, result = []

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.15.1.2.Example

Define the return value-empty list

result = []

Plugin-in name

sCmdName = 'FT_Plugin'

Plug-in instructions and parameters

sParams = ['FT_reset']

Send the plugin command

nRet = cps.HRIF_HRAppCmd(0,0,sCmdName, sParams, result)

3.15.2.HRIF_WriteEndHoldingRegisters

3.15.2.1.Description: Write the end-connected Modbus slave register.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
nSlaveID	Slave station ID	int	0/1	Slave station ID
nFunction	Function code	int	>0	Function code: 0x01-Read the coil register 0x02-Read the line with a discrete input register 0x03-Read-hold register 0x04-Read on the input register 0x05-Write a single coil register 0x06-Write a single hold register 0x0f-Write the multiple coil registers 0x10-Write the multiple hold registers
nRegAddr	Register address	int	>0	Register start address
nRegCount	Number of registers	int	>0	Number of registers
nData	Register data	list[int]	-	Register data

* Note: End at EtherCAT bus version IO.

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.15.2.2.Example

```
# Set the slave station ID=1
nSlaveID = 1

# Set the function code
nFunction = 16

# Set the register start address
nRegAddr = 100

# Set the register number
nRegCount = 2
```

#Set up the register data

nData=[196,34465]

nRet = cps.HRIF_WriteEndHoldingRegisters(0,0,nSlaveID, nFunction, nRegAddr, nRegCount, nData)

3.15.3.HRIF_ReadEndHoldingRegisters

3.15.3.1.Description: Read the Modbus slave register.

✓ Input variable

Input variable	Name	Data type	Valid range	Content
boxID	Electricity box ID	int	0~5	Box ID number, default value =0
rbtID	robot ID	int	0~5	Robot ID number, default value =0
result	Return value	list		Pass into the empty list, result = []
nSlaveID	从站 ID	int	0/1	从站 ID
nFunction	Function code	int	>0	Function code: 0x01-Read the coil register 0x02-Read the line with a discrete input register 0x03-Read-hold register 0x04-Read on the input register 0x05-Write a single coil register 0x06-Write a single hold register 0x0f-Write the multiple coil registers 0x10-Write the multiple hold registers
nRegAddr	Register address	int	>0	Register start address
nRegCount	Number of registers	int	>0	Number of registers

* Note: End at EtherCAT bus version IO.

✓ Output variable

Output variable	Name	Data type	Valid range	Content
result	Register data	list[int]	-	Register data

✓ Return value

Return value	Name	Data type	Valid range	Content
nRet	Return value	int	An integer value > 0	nRet=0 : Returns that the function call was successful nRet>0 : Returns the error code of the call failure

3.15.3.2.Example

Set the slave station ID=1

nSlaveID = 1

Set the function code

nFunction = 3

Set the register start address


```
nRegAddr = 100
```

```
# Set the register number
```

```
nRegCount = 4
```

```
# Define the return value-empty list
```

```
result = [ ]
```

```
# Read the register data
```

```
nRet = cps.HRIF_ReadEndHoldingRegisters(0,0,nSlaveID, nFunction, nRegAddr, nRegCount, nRegCount,  
result)
```

```
for i in range(len(result)):
```

```
    print(result[i])
```

Chapter IV Appendix

4.1.Reference documents

1. Please refer to the error code and status machine description [《HansRobot ErrorCode.docx》](#).
2. For the use of communication protocols for Han's robots, please refer to [《HanRobotV5 Control Communication Protocol Interface》](#).
3. For the use of the C # SDK for Han's robots, please refer to [《HanRobot Library C#》](#).
4. For the use of the C # SDK for Han's robots, please refer to [《HanRobot Library Java》](#).
5. For the use of the C # SDK for Han's robots, please refer to [《HanRobot Library C++》](#).